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Revisiting the trading activity of high-frequency trading firms around ultra-fast flash events

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Abstract

We investigate high-frequency traders' behavior in the context of the fastest and most extreme price movements (EPMs) that can be observed in the market, specifically ultra-fast flash events, challenging the methodologies employed in the academic and practitioner literature for identifying sudden liquidity black holes. To refine the price-shock identification methodology, we introduce a new approach called sequence-based flash events (SFEs), which relies on tick sequences instead of predetermined fixed-time intervals within which all flash events in the sample are assumed to occur. This alternative methodology offers the advantage of pinpointing the exact time and duration of a crash, which, in turn, provides a way to more accurately define the observation windows around it. We compare our sample of SFEs with both the so-called "mini flash crashes", as identified by the Nanex detection algorithm, and the so-called EPMs, as identified by Brogaard et al. (2018). We use close and open prices, as well as high and low prices. Based on our sample of SFEs, we find no evidence that HFTs trigger extreme price shocks. However, we find that HFTs exacerbate SFEs by increasing the net imbalance in the direction of these shocks as they occur. Finally, we show that the choice of the price-shock identification methodology is critical. Thus, we urge regulators to exercise caution and avoid hasty conclusions regarding HFTs' contribution to price stability in stressful market conditions.

Keywords: Flash events, Mini flash crashes, Extreme price movements, High-frequency trading, Trading activity, Liquidity, Market stability, Market microstructure

JEL Classification: G1, G10, G14

Introduction

This study introduces a novel methodology aimed at both identifying extreme market shocks with greater precision and analyzing the role played by high-frequency traders (HFTs) before, during, and after these events. We focus particularly on ultra-fast flash events, a subcategory of "flash events" that incorporates so-called "mini flash crashes" (MFCs) and "extreme price movements" (EPMs). Our aim is to comprehensively explore these events, integrating both industry and academic perspectives, to provide deeper

insights into the dynamics of financial markets and HFTs' contribution to market (in) stability.

Practitioners refer to the fastest and most extreme price movements as “ultrafast extreme events” (Johnson et al. 2013), “flash equity failures,” or “MFCs” (Nanex LLC, Nanex 2010). They represent sudden and significant changes in trade prices that adhere to predefined rules. Specifically, an MFC corresponds to a price change exceeding 0.8%, accompanied by at least ten consecutive tick movements in the same direction before a change in the opposite direction occurs, all within a duration of 1.5 s (Nanex LLC, Nanex 2010).

Academically, these events are referred to as EPMs and correspond to extremely significant shifts in the National Best Bid and Offer (NBBO) midquotes (Brogaard et al. 2018). They represent multiple sequential trades and are retrospectively identified based on a statistical approach, specifically utilizing the 99.9th percentile of the absolute return distribution.

Despite similarities, MFCs and EPMs differ notably in terms of identification methods, duration, and statistical properties. For example, there are differences in their identification methodologies (ex-ante for MFCs versus ex-post for EPMs), their duration (10-s time intervals for EPMs versus 1.5 s time intervals for MFCs), and descriptive statistics (lower mean return for EPMs due to filtering mechanisms). Consequently, MFCs appear to be more extreme than EPMs, despite the latter's name. This prompts a unified approach.

Our study adopts a systematic approach, scrutinizing ultra-fast flash events through various methodologies, including a novel one proposed herein. This aspect is critical, as existing studies, typically constrained to using a single methodology, have produced contradictory results regarding HFTs' behavior during these events. Our study aims to discern whether these discrepancies stem from the datasets used or the methodologies employed. We therefore test whether these divergent perspectives are reconcilable, analyzing HFT behavior on the same stocks and over the same period.

The research process involves three key steps: (1) replicating practitioner-focused MFCs and academic-focused EPMs, (2) introducing a new “sequence-based flash events” (SFEs) methodology, (3) comparing SFEs with both practitioner-focused MFCs and academic-focused EPMs. Our novel methodology distinguishes itself by capturing flash events without relying on fixed-time windows, instead utilizing tick sequences to accurately identify the duration of each flash event and define precise observation windows. Our data include tick-by-tick data, millisecond-level timestamps, and a flag indicating whether the liquidity provider or consumer is an HFT or a non-high-frequency trader (NHFT).

This study contributes to the field of finance by systematically comparing different price-shock identification methodologies, including a more precise methodology that we detail in the next section. To the best of our knowledge, this is the first study to highlight the inconsistency of findings across methods. Interestingly, we find no evidence that HFTs trigger extreme price shocks. However, we also find a discernible destabilizing effect during crashes. Our findings also highlight the methodology-dependent nature of conclusions drawn about HFTs' contribution to market stability. Thus, we caution against premature regulatory judgments and advocate for further research in this area.

The remainder of the paper is organized as follows: Section 2 presents a graphical and intuitive analysis of flash events to facilitate understanding of our identification methodology. Section 3 reviews the relevant literature on flash events, MFCs, and EPMs, further delineating the contribution of our study. Section 4 introduces the data, presents the different identification methods we use in the study, summarizes the statistics, and defines the different variables. Section 5 is dedicated to the empirical analysis and presents our results. Section 6 summarizes the different robustness checks performed and addresses the limitations of the study. Section 7 concludes.

A visual illustration

MFCs are defined as abrupt, extreme, and brief price fluctuations that involve a minimum of ten tick movements in a consistent direction, followed by a movement in the opposite direction, partially or fully correcting themselves within a few seconds. These MFCs can be considered as scaled-down versions of flash crashes. As highlighted by Johnson et al. (2013), they differ from flash crashes in two fundamental respects. First, MFCs have a considerably shorter duration, typically lasting only a few seconds, in contrast to flash crashes that can persist for several minutes. This abbreviated timeframe restricts the opportunity for human intervention to effectively address the crash. Second, the remarkably swift speed of occurrence and recovery observed in most MFCs indicates that they are unlikely to be triggered by the arrival of exogenous news.

To illustrate the concept, Fig. 1 portrays an instance of a downward MFC that impacted Apple (AAPL) stock on January 22, 2008. This crash unfolded within a remarkably short time span of 1,180 ms (starting at 10:02:24.100 and ending at 10:02:25.280). The transaction price experienced a rapid decline of 5.06% within this compressed interval before promptly rebounding to its initial level.

As defined in Johnson et al. (2013) and Nanex (2010), the aforementioned MFCs are filtered out based on three rules. There must be a price change exceeding 0.8%, with at least ten tick movements in the same direction before ticking in the other direction. This price change must happen within 1.5 s.

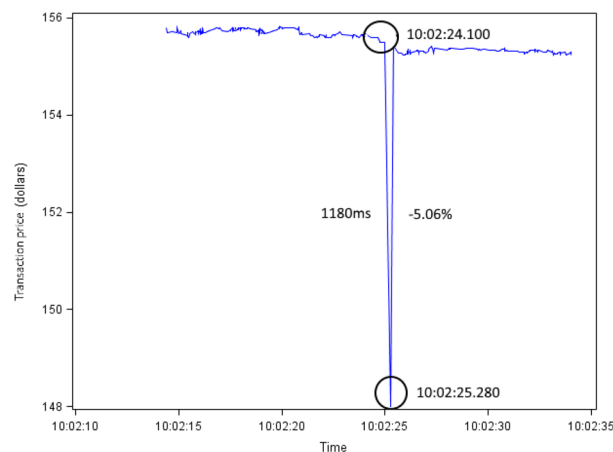


Fig. 1 A down MFC on AAPL on January 22, 2008. Source: Tickdata

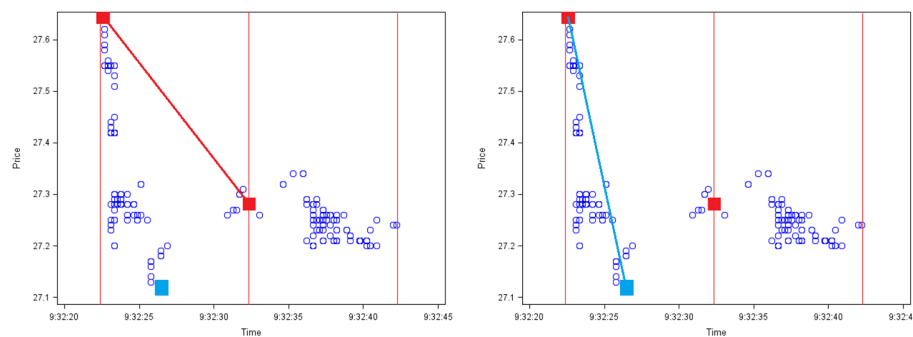


Fig. 2 Measured crash versus true crash on AA - January 22, 2008. Source: Tickdata

Brogaard et al. (2018) instead focus on EPMs, which are computed as returns above the 99.9th percentile of the empirical distribution of absolute returns. In total, they identify 45,200 EPMs at 10 s intervals on 40 large stocks in the Nasdaq HFT database over the period from 2008 to 2010. The median absolute return of an EPM is 0.436% at pre-specified 10 s time intervals.

Both methodologies could be questioned. First, we might question whether (Brogaard et al. 2018) really captured EPMs because they used fixed-time windows and potentially never identified the peaks and troughs of the price movements. This is particularly important, as price movements can be extremely short-lived (a few milliseconds). Even when they changed the time interval from 1 s, 5 s, 10 s, and 30 s to 1 min, there is no guarantee that they identified the peaks and troughs of a shock due to the fixed time window. In addition, the shorter the time interval, the more critical it is to identify the peaks and troughs of a flash event, as intervention could occur in the next fixed-time interval.

To better understand why it may be misleading, let us consider a downward crash and examine two different undesirable elements. First, when a larger drop occurs *within* an interval (larger than the drop between the opening and closing prices of the interval), (Brogaard et al. (2018)) underestimate the downward crash and include some price correction in their EPM. In other words, resiliency may already be occurring within the EPM interval. Second, when a larger drop occurs *directly after* the closing price of the interval, Brogaard et al. also underestimate the downward crash and distort the next interval because the crash has not yet ended.

Figure 2 offers a visual illustration of why measuring EPMs from open to close may be misleading when fixed-time intervals are used and determined ex-ante. We consider the first undesirable element explained above, where the EPM is underestimated, and the fixed-time interval includes some price correction.

The graph shows an EPM occurring in Alcoa (AA) stock on January 22, 2008. The red vertical lines represent the 95th, 96th, and 97th 10 s intervals for that day. Each dot stands for transactions within the same millisecond.¹ The red square dots represent the opening and closing transaction prices within the predetermined 10 s interval (from interval 95 to 96), while the light-blue dot represents the true extreme transaction price

¹ When two or more transactions occur within the same millisecond, all the transactions graphically appear as the same dot.

of the sequence. The measured crash, represented by the red diagonal line (Fig. 2 on the left), underestimates the true crash, represented by the light-blue diagonal line (Fig. 2 on the right). This illustrates that the open-close EPM approach (i.e., measuring the magnitude of the crash based on the closing price and opening price of the interval) is not well suited, especially when fixed-time intervals are used to identify EPMs.

Second, we could also question Nanex's identification methodology, which appears arbitrary. For example, a price change of 90% happening in less than 10 *consecutive tick* movements *in the same direction* would not qualify as an MFC, while a price change slightly exceeding 0.8% in 10 consecutive ticks would. Similarly, a price change of 90% happening in slightly more than 1.5 s would not qualify as an MFC, while a price change slightly exceeding 0.8% in less than 1.5 s would.

In this study, we aim to alleviate some of the drawbacks of these two approaches to better capture flash events. The innovation relies on the fact that we first identify the extreme price shock, which we call an SFE, and then redefine the window around it without relying on any predetermined fixed-time interval for the identification of the shock. Our methodology entails thinking in terms of sequences rather than intervals. Thus, we rely on the equity curve. As long as the equity curve does not reverse, we stay in the same sequence. Once the equity curve changes direction, we then move to the next sequence. Figure 3 offers a visual illustration of how SFEs work.

We first measure the tick curve by summing ticks between the maximum and minimum (and vice versa for upward SFEs). To capture the full extent of the shock, we continue measuring the equity curve *in the following interval* as long as the equity curve does not move in the direction opposite to that of the previous equity curve. The first sequence goes from point 1 to point 2 in Fig. 3. To ease comparison across methodologies and focus solely on ultra-fast flash events, the duration of a sequence in our SFEs does not exceed 1.5 s. We also require a 100% retracement in the second sequence (from

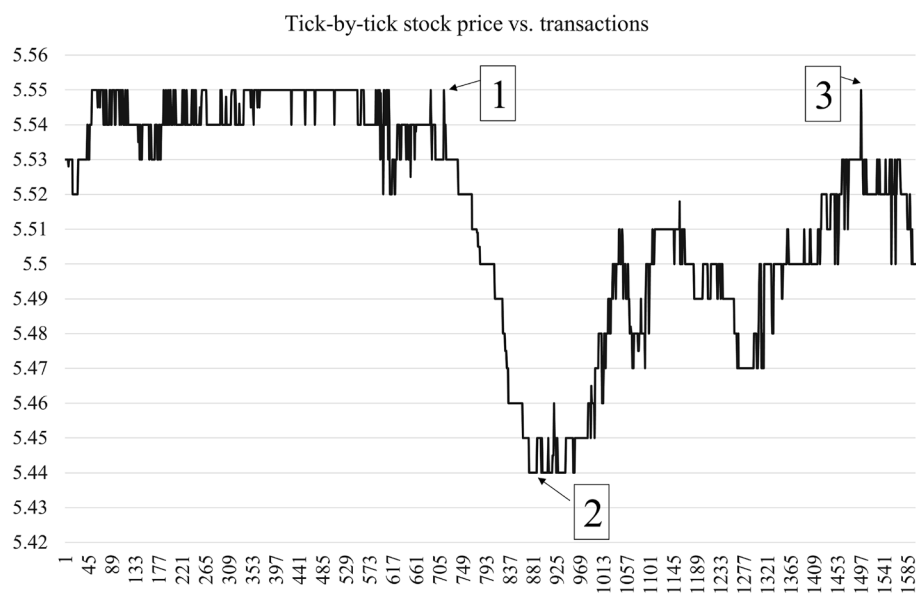


Fig. 3 Example of an SFE. Source: Tickdata. The number on the X axis corresponds to ticks since the open of the day

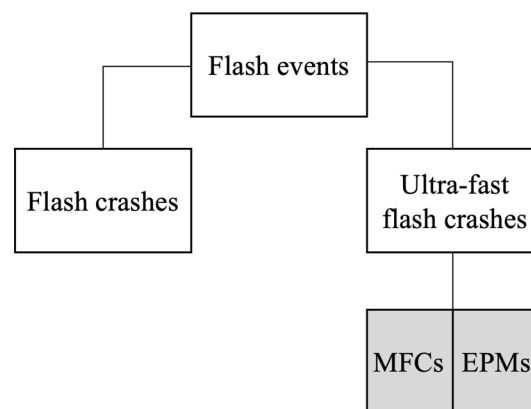


Fig. 4 Decomposition of flash events

point 2 to point 3 in Fig. 3) to include only liquidity-driven flash events and exclude information-driven shocks.

Brief literature review

In this section, we introduce the relevant literature, focusing on identifying and measuring ultra-fast flash events, the impact of HFTs on the market, and the role played by HFTs during flash and ultra-fast flash events.

Identifying and measuring ultra-fast flash events

Our study focuses on “ultra-fast flash crashes”, which are shorter in duration than “flash crashes”, as indicated in Fig. 4, although both belong to the category of “flash events”.

Within the body of the literature focusing on ultra-fast flash events, these events are referred to as either MFCs², when they meet (at least partially) the tick, time, and price-change criteria defined by Nanex (2010), or EPMs when they meet the price-change criterion defined by Brogaard et al. (2018).

According to Nanex (2010), an MFC must include at least 10 tick movements in the same direction before ticking in the other direction, all within 1.5 s, and with an absolute price change exceeding 0.8%. The tick criterion is consistently set at 10 ticks in the literature, as seen in Golub et al. (2012), Johnson et al. (2013), and Ozenbas and Schwartz (2018). The duration criterion is typically set at 1.5 s (Nanex 2010) but is sometimes extended, as in Braun et al. (2018), Félez-Vinas (2018), or Laly and Petitjean (2020). In addition, Félez-Vinas (2018) adds a recovery criterion in her identification of MFCs, stating that a stock must recover at least 90% of the initial price change within 300 s.

According to Brogaard et al. (2018), an EPM is identified as a 10 s interval whose return belongs to the 99.9th percentile of the absolute return distribution. This 10 s interval has been chosen to capture brief liquidity dislocations that are sufficiently relevant in terms of magnitude.

To the best of our knowledge, MFCs were first mentioned in the literature by Johnson et al. (2013), who reported 18,520 MFCs in the United States (U.S.) equity markets

² MFCs are also sometimes called “down and up crashes” as in Golub et al. (2012), “black-swan crashes and dashes” or “ultrafast extreme events” as in Johnson et al. (2013).

alone between January 3, 2006 and February 3, 2011, i.e., approximately 15 MFCs per day. Félez-Vinas (2018) reported 2,105 and 947 MFCs on the Spanish IBEX35 and Italian FTSE MIB exchanges, respectively, over the period from November 2012 to April 2013, i.e., approximately 8 and 17 MFCs per day, respectively. Both studies reveal that MFCs occur globally and not only in the U.S. Regarding EPMs, they were first mentioned in the literature by Longin (2000), who described them as either market corrections occurring during ordinary periods or crashes occurring during extraordinary periods. However, it is Brogaard et al. (2018) who placed EPMs in the context of high-frequency trading, determining the price-change rule to capture these “ultra-fast flash crashes” ex-post.

Finally, Aquilina et al. (2018) identify extreme events that they call “mini flash crashes/rallies” in the United Kingdom (UK) equity market over the period from January 2014 to June 2015. In contrast to Nanex (2010), Aquilina et al. (2018) define “mini flash crashes” as large price movements that revert within a short time window during which there is a high level of traded volume. They identify a total of 40 flash events whose drop or spike duration can last up to 10 minutes. In contrast to Aquilina et al. (2018), we believe that these intraday market shocks are traditional flash crashes and do not belong to the universe of ultra-fast liquidity black holes that we study here. For a complementary review of the literature on MFCs and EPMs, see Laly and Petitjean (2020) and Steffen (2023).

The impact of HFTs on the market

Many benefits have been associated with the arrival of HFTs in electronic markets, such as a decrease in spreads (Stoll 2014; Boehmer et al. 2021), a decrease in transaction costs (Jones 2013), a decrease in short-term volatility (Hagströmer and Nordén 2013; Hasbrouck and Saar 2013), as well as an improvement in price efficiency (Carrion 2013; Brogaard et al. 2014). Consequently, some of the trading strategies used by HFTs, such as statistical arbitrage, directional trading, and market-making strategies, are perceived as mostly beneficial in the literature (Zaharudin et al. 2022). However, some major drawbacks have also been associated with the presence of HFTs, the most problematic of which is the suspicion that HFTs may harm modern automated markets (Hasbrouck and Saar 2013), thus generating externalities related to market quality, liquidity, and volatility.

Regarding HFTs’ negative impact on market quality and liquidity, Hirschey (2021) provides evidence that HFTs identify patterns in past trades and orders that allow them to anticipate and trade ahead of the order flow of other investors (front-running), thus increasing trading costs for non-HFT (NHFT) participants. Similarly, Goldstein et al. (2023) find that HFTs impose a welfare externality by selectively crowding out the most profitable limit orders on the Australian Securities Exchange. The authors show that HFTs strategically use limit-order book information to supply liquidity on the thick side of the order book and demand liquidity from the thin side, especially during periods of high volatility and when trading speed increases. This observation echoes the findings of both Menkveld (2014), who argues that the migration to electronic trading coincides with the occasional disappearance of liquidity supply, and Menkveld and Zoican (2017), who claim that a faster exchange does not necessarily improve liquidity, especially when price quotes are met with “speculative high-frequency bandits”. Finally, Abreu (2022) shows that while HFTs do improve market liquidity on average, they do so at the expense

of NHFTs. In other words, HFTs are the primary beneficiaries of this enhanced market liquidity, while other market participants must contend with worsened liquidity levels.

Regarding HFTs' negative impact on volatility, Breckenfelder (2019) finds that the competition among HFTs increases intraday volatility. Moreover, the author shows that an increase in HFT competition increases HFTs' speculative behavior, reduces liquidity, and thus increases short-term volatility (implicitly leading to more frequent intraday price shocks). Similarly, Boehmer et al. (2018) show that HFTs have highly correlated trading strategies and a dominating presence, so their actions can exacerbate price movements and increase short-term volatility. This is especially problematic due to the fact that, as explained by Goldstein et al. (2014), "errant or poorly designed HFT programs without necessary risk controls can lead to occasional shocks or disruptive events that affect markets and, perhaps as importantly, the general public's perception of them" (p. 198). Finally, Shkilko and Sokolov (2020), in their study of a series of exogenous weather episodes (rain or snow) that temporarily remove the speed advantage of the fastest traders (by disrupting their microwave networks), find that higher HFT participation during periods without precipitation is associated with higher volatility levels. Again, these findings imply that increased HFT activity in the market correlates with a greater likelihood of price shocks. Overall, while some trading strategies used by HFTs are generally considered beneficial, others are unequivocally harmful. These harmful strategies include front-running, order anticipation, quote matching, spoofing, layering, and quote stuffing (Zaharudin et al. 2022).

Role played by HFTs during flash events and ultra-fast flash events

On a microstructural level, HFTs are especially suspected of exhibiting harmful behavior during sudden intraday shocks, as pointed out by Bellia et al. (2022), who show that electronic designated market makers consume liquidity when it is most needed. As such, there seems to exist a trade-off between the greater liquidity and efficiency provided by market-making HFTs in normal times and the disruptive consequences of their quoting and trading activity in times of market stress. It is important, at this stage, to emphasize that intraday market shocks can take one of two forms: the form of "flash crashes", which can be defined as sudden and extreme (up or down) price changes that partially or totally self-correct within a maximum of a few dozen minutes, and the form of "ultra-fast flash crashes" represented by "MFCs" and "EPMs"—sometimes called "EPMs with reversals" as in Steffen (2023)—which are faster versions of "flash crashes" that partially or totally self-correct within only a few seconds. Many such flash crashes have occurred in the past decade, whether in equities (2010, 2015, and 2023 U.S. equity flash crashes), in fixed income (2014 Japanese government-bond flash crash, 2014 U.S. government-bond flash crash, 2015 German-bond flash crash, and 2018 Italian-bond flash crash), in currencies (2015 USD flash crash and 2016 GBP flash crash), in commodities (as reported by Mas-sad (2015)),³ or, more recently, in cryptocurrencies. Despite the implementation of various protective measures, such as limit-up/limit-down bands and circuit-breaker updates in the U.S. stock market, flash crashes continue to occur and are expected to persist in

³ A good example is the event of May 4, 2023, where the DJIA closed down nearly 300 points following West Texas Intermediate (WTI) futures tumbling 7.3% shortly after trading opened on the New York Mercantile Exchange.

the future. This is evidenced by the flash crash that occurred on January 24, 2023, during which highly liquid U.S. stocks experienced a significant drop of approximately 10% at the market open, followed by a rapid recovery, in spite of the existing safeguards.

Flash events seem to have become more frequent with the emergence of high-frequency trading. For example, Leal et al. (2016) find that they do not occur when the market (in their agent-based model) is populated with low-frequency traders only. While some studies have examined the role played by HFTs during flash events (Kirilenko et al. 2017; Aldrich et al. 2017; Menkveld and Yueshen 2019), the literature regarding the role played by HFTs during the fastest and most extreme price shocks (during ultra-fast flash events) is much more limited.

Within the literature focusing on the famous May 6, 2010, Flash Crash, findings regarding the role played by HFTs point to similar conclusions. In their study of HFTs' behavior in the E-mini S&P500 futures market during the Flash Crash, Kirilenko et al. (2017) find that HFTs did not cause the crash. The large automated selling program of a mutual fund later identified as Waddell & Reed did. They also find that HFTs did not fundamentally change their trading pattern during the crash. However, the authors note that just before the market was paused for 5 s, HFTs liquidated 2,000 contracts accumulated earlier (in an already illiquid market), coinciding with a significant additional price decline, i.e., the most abrupt price decline of the crash. In contrast, traditional market makers (NHFTs by extension) did not liquidate their accumulated inventory. In that sense, HFTs contributed to the extent of the Flash Crash. Their findings are consistent with the empirical study led by Aldrich et al. (2017). As for Menkveld and Yueshen (2019), they conclude that the crash cannot be attributed to the mutual fund alone and that it is rather the result of the interaction between market participants that degenerated into a flash crash. This interpretation is consistent with the findings of Demirer et al. (2019), who conclude that "the market switched to a herding state just before the crash and remained in this state over the duration and aftermath of the crash". In this case, the authors establish a clear link between flash events and herding among market participants. However, nothing more is said about the role played by HFTs.

In a more recent working paper focusing on flash crashes, Bellia et al. (2022) use the econometric "drift burst" approach of Christensen et al. (2022) to identify 148 flash crashes on 37 liquid French stocks belonging to the CAC40 index and traded on NYSE-Euronext Paris in 2013. The "drift burst" hypothesis postulates that there exist short-lived, locally explosive trends in the price paths of financial assets (in other words, periods when the price is trending sharply) as characterized by flash events. They investigate the behavior of different groups of traders during the identified flash crashes and find that designated market makers (DMMs), most of them HFTs, do not provide sufficient liquidity to avoid flash crashes occurring in isolated stocks and that they even consume liquidity when crashes simultaneously affect several stocks. Moreover, they find that DMMs do not mitigate the effects of illiquidity shocks during flash crashes, which somewhat contradicts the findings of Anand and Venkataraman (2016), Clark-Joseph et al. (2017), and Bessembinder et al. (2020). Finally, the authors show that in most cases, flash crashes are originated by either investment banks' proprietary trading or by client trading from investment banks via sponsored access. The crashes are then amplified by

the reaction of investment bank DMMs, in particular, when several stocks are simultaneously impacted.

Finally, the literature focusing on ultra-fast flash events points to different conclusions because, as pointed out by Steffen (2023), there is no consensus regarding the role played by HFTs during “ultra-fast flash crashes”. On the one hand, Brogaard et al. (2018), in their empirical study of EPMs, find little evidence of HFTs causing EPMs. Indeed, the authors conclude that HFTs are net suppliers of liquidity to NHFTs during EPMs on single stocks and that HFTs do supply liquidity during the most extreme EPMs. However, they also concede that when several stocks experience simultaneous EPMs, HFT liquidity demand then dominates their supply, so that HFTs de facto become net demanders of liquidity. On the other hand, the rare studies focusing on MFCs are more skeptical about the role played by HFTs during these liquidity black holes and speculate that HFTs may have a disruptive influence. For example, Johnson et al. (2013) argue that their findings are consistent with the emergence of “crowds of predatory algorithms” that destabilize the market, and Golub et al. (2012) postulate that “given the speed and the magnitude of the crashes, it appears likely that Mini Flash Crashes are caused by HFT activity”, which echoes Gao and Mizrach (2016), who find that HFT-trading activity raises the probability of market breaks. While HFTs are portrayed as exhibiting beneficial behavior during EPMs, they tend to be portrayed as exhibiting detrimental behavior during MFCs. It is thus crucial to test whether these conflicting views are reconcilable.

Data

Data and sample

We obtain data from both Tickdata and Nasdaq and work with both datasets together. On the one hand, we use tick-by-tick data timestamped to the millisecond on the NBBO from Tickdata for 74 stocks.⁴ The data from Tickdata account for all transactions on U.S. stock exchanges⁵ for the 74 stocks at our disposal. On the other hand, we use tick-by-tick data timestamped to the millisecond on trades from Nasdaq OMX for the same 74 stocks, as compared to 40 stocks in Brogaard et al. (2018). This dataset is the so-called Nasdaq HFT dataset provided to academics under a non-disclosure agreement. The dataset is the same as the one used in O’Hara et al. (2014), Carrion (2013), Gerig (2015), Brogaard et al. (2017, 2018), and Hirschey (2021). It includes a flag that enables us to know if the liquidity consumer/supplier is an HFT or an NHFT. A code (HH, HN, NH, or NN) is associated with each transaction on Nasdaq. This flag is not provided by the other stock exchanges. The first letter of the code refers to the liquidity seeker, and the second letter refers to the liquidity supplier. For example, when a transaction occurs between an HFT and an NHFT, the code can either be HN or NH, depending on who is on the liquidity-demand side and who is on the liquidity-supply side. HN indicates that the HFT demands liquidity while the NHFT supplies liquidity. Conversely, NH indicates that the NHFT demands liquidity while the HFT supplies liquidity.

⁴ Table 11 in the Appendix reports the ticker, company name, market cap, and sector (GICS classification) of these stocks. This list includes 32 large-cap, 30 medium-cap, and 12 small-cap stocks.

⁵ NYSE American (formerly AMEX and NYSE MKT), NASDAQ OMX BX (Boston), National Stock Exchange (Cincinnati), ISE (International Securities Exchange), DirectEdge A, DirectEdge X, Chicago, NYSE, NYSE Arca, NASDAQ, CBOE, NASDAQ OMX PSX (Philadelphia), BATS Y-Exchanges, and BATS.

We proceed with these two datasets as follows. First, we use the aggregated data from Tickdata to identify price shocks as they originate from successive transactions that are likely to occur on different exchanges. Second, we use the Nasdaq HFT dataset to observe HFTs' behavior around and during these shocks. Nasdaq is by far the U.S. exchange with the highest market share (36%) based on the number of trades for the 74 stocks included in our sample, which makes a focus on Nasdaq all the more relevant. Third, in terms of window-sizing around a shock, we use a window of 30 s around each extreme price shock, i.e., ten 1.5 s intervals before and after the shock. Fourth, we use transaction prices in identifying MFCs and SFEs in order to consider the full magnitude of each crash (from top to bottom or from bottom to top). However, we still use mid-quotes in identifying EPMs to be consistent with Brogaard et al. (2018). Finally, we focus on trading activity from 9:30 a.m. to 4:00 p.m. ET in order to consider the full trading period from the opening to the closing of the U.S. equity market.

Identifying flash events

We begin by revisiting the first methodology proposed by Nanex (2010), which is used to identify MFCs. According to Nanex (2010), an MFC is characterized by at least 10 consecutive tick movements in the same direction (excluding trades with no tick change) occurring within a duration of 1.5 s and involving a price shock exceeding 0.8%. This procedure is also followed by Golub et al. (2012) and Johnson et al. (2013). However, the selection criteria, such as the 0.8% absolute return threshold, are arbitrarily defined.

To address this fact, we propose an alternative identification methodology based on SFEs. In this approach, we eliminate the price-change rule and consider all sequences, regardless of their return, before ranking them using percentiles. As with the EPMs defined by Brogaard et al. (2018), we keep the most extreme sequences with an absolute log return surpassing the 99.9th percentile of the absolute log return distribution. Unlike Brogaard et al. (2018), however, we still incorporate the tick rule from Nanex (2010) to capture fast and abrupt shocks. As long as the equity (or tick) curve remains in the same direction, we consider it as part of the same sequence.

Importantly, our SFEs are exclusively determined by their duration and are not constrained by predetermined fixed-time intervals. This is not the case for EPMs, which are based on returns from open to close (OC) prices and defined within fixed-time intervals established at the start of the day. Consequently, these OC-EPMs may not accurately represent the true magnitude of a price shock, as demonstrated in Section 2. To better capture the magnitude of these EPMs, we propose to use the high and low (HL) prices observed during the time window. We refer to these modified EPMs as "HL-EPMs" in the subsequent analysis. Unlike EPMs based solely on open and close prices, HL-EPMs capture the true amplitude of a shock while still being within the same fixed-time interval. It is important to note that even HL-EPMs may not fully capture the entire magnitude of a shock if it spans two time intervals. Such an issue never arises in our sample of SFEs.

In our time-series analysis, we compare all methodologies (MFCs, SFEs, OC-EPMs, and HL-EPMs) by using a duration of 1.5 s for the windows of observation around the flash events.⁶

⁶ See Section 5.5 for the robustness checks. Among others, we explore alternative time intervals of 1 s, following (Brogaard et al. 2018), and 2 s, following Féliz-Vinas (2018), with similar conclusions.

Variables

We present below how we compute the main variables of the study. These variables are either trading-activity measures, HFT-participation measures, liquidity measures, or net HFT-activity measures.

Liquidity measures

We measure order book liquidity through four different metrics: depth at best prices (*Depth*), dollar depth at best prices (*Dollar Depth*), bid-ask spread (*Quoted Spread*), and relative bid-ask spread (*Relative Spread*).

We define *Relative Spread* as follows:

$$\text{Relative Spread}_{i,t} = (\text{Best Ask Price}_{i,t} - \text{Best Bid Price}_{i,t}) / \text{Mid quote}_{i,t} \quad (4.1)$$

where $\text{Best Bid Price}_{i,t}$ and $\text{Best Ask Price}_{i,t}$ represent the best bid and ask prices of the order book, respectively, i represents the stock, and t represents the time of the quote update. The Midquote is defined as the average of the best bid and best ask prices.

Trading-activity measures

We measure trading activity through three different metrics: number of trades (*Total Trades*), dollar volume (*Dollar Volume*), and share volume (*Share Volume*).

The first metric, *Total Trades*, measures the total number of trades per stock per interval (<1.5 s in Panel B and =1.5 s in Panels A and C of Table 1 as well as in Panels A, D, and E of Table 2). The second metric, *Dollar Volume*, measures the average dollar volume per stock per interval. The third metric, *Share Volume*, measures the average share volume per stock per interval.

HFT-participation measures

We measure HFTs' participation through three different metrics: proportion of HFT trades (*Proportion HFT Trades*), proportion of HFT shares (*Proportion HFT Shares*), and proportion of HFT volume (*Proportion HFT Volume*), following Carrion (2013).

First, we define *Proportion of HFT Trades* as follows:

$$\text{ProportionHFTtrades}_{i,t} = (HH_{i,t}) / (HH_{i,t} + NN_{i,t}), \quad (4.2)$$

based on Nasdaq trades.

Second, we define *Proportion of HFT Shares* as follows:

$$\text{ProportionHFTshares}_{i,t} = (HH_{i,t}) / (HH_{i,t} + NN_{i,t}), \quad (4.3)$$

based on Nasdaq shares traded.

And third, we define *Proportion of HFT Volume* as follows:

$$\text{ProportionHFTvolume}_{i,t} = (HH_{i,t}) / (HH_{i,t} + NN_{i,t}), \quad (4.4)$$

based on Nasdaq dollar volume.

Table 1 Summary statistics for the Full, MFC, and SFE samples

Panel A: Full sample					
	Minimum	Median	Mean	Maximum	Std Dev
Absolute log return, %	0	0	0.0119	19.64	0.0335
Total trades (all U.S. exchanges)	0	0	3.76	1,205	11.25
Total trades (Nasdaq)	0	0	1.57	880	5.45
Proportion HFT trades (Nasdaq)	0	0	0.1821	1	0.3611
Proportion HFT shares (Nasdaq)	0	0	0.1784	1	0.3591
Proportion HFT volume (Nasdaq)	0	0	0.1784	1	0.3591
Share volume	0	0	850.36	28,368,232	10,720.59
Dollar volume	0	0	36,099.82	1,458,100,000	459,793.2
Depth	2	6	23.77	100,816	100.00
Dollar Depth	6.34	152.75	365.57	2,225,817	1819.76
Quoted spread, \$	0.01	0.02	0.0412	113.91	0.1143
Relative spread, %	0.0022	0.0631	0.1225	56.21	0.7775
Panel B: MFC sample					
	Minimum	Median	Mean	Maximum	Std Dev
Crash duration, ms (MFC specific)	0	60	216	1,487	338
Total tick change (MFC specific)	10	12	13.43	30	3.78
Absolute log return, %	0	0.2167	0.5577***	19.36	1.6214
Total trades (all U.S. exchanges)	3	64	99.21***	883	109.01
Total trades (Nasdaq)	0	25	44.64***	672	60.32
Proportion HFT trades (Nasdaq)	0	0.6667	0.6195***	1	0.2824
Proportion HFT shares (Nasdaq)	0	0.5667	0.5492***	1	0.2955
Proportion HFT volume (Nasdaq)	0	0.5661	0.5493***	1	0.2954
Share volume	500	19,200	55,008***	3,138,737	175,315
Dollar volume	13,343	634,635	2,320,004***	146,020,000	9,855,164
Depth	2	7.30	28.62	1,505	100.79
Dollar depth	20.42	176.61	581.25	36,062.53	2,554.32
Quoted spread, \$	0.01	0.0279	0.9129**	113.91	8.19
Relative spread, %	0.0093	0.0884	0.4999**	42.89	2.99
Panel C: SFE sample					
	Minimum	Median	Mean	Maximum	Std Dev
Absolute log return, %	0	0	0.63***	24.11	38.77
Total trades (all U.S. exchanges)	11	17	20.66***	310	13.06
Total trades (Nasdaq)	0	5	7.06***	128	8.38
Proportion HFT trades (Nasdaq)	0	0.26	0.33***	1	0.31
Proportion HFT shares (Nasdaq)	0	0.21	0.32***	1	0.32
Proportion HFT volume (Nasdaq)	0	0.21	0.32***	1	0.32
Share volume	1,110	4,205	8731.84***	3,947,060	66,156.09
Dollar volume	9,805	112,098	384,007.01***	638,279,950.20	10,273,938.23
Depth	2	38.40	84.76	2,701.92	148
Dollar depth	24.85	912.34	1,826.70	94,057.33	3,576.94
Quoted spread, \$	0.01	0.01	0.19**	124.83	4.12

Table 1 (continued)**Panel C: SFE sample**

	Minimum	Median	Mean	Maximum	Std Dev
Relative spread, %	0.01	0.05	0.13**	44.81	1.48

The table reports descriptive statistics for the full sample (Panel A), the sample of MFCs (Panel B), which follows Nanex identification methodology, and the sample of SFEs (Panel C), which is the alternative methodology we propose. N corresponds to the number of time intervals in the first regression model. The lower the number of ultra-fast flash events, the lower N . We use a sub-1.5 s variable interval for Panel B and a fixed 1.5 s interval for Panels A and C. All data are from Tickdata except Total trades and Proportion of HFT trades, HFT shares and HFT volume which are from Nasdaq. The mean of Absolute log return, Total trades, Depth, Dollar volume, Share volume, Quoted spread and Relative spread is computed in two steps. First, we compute the P50 by stock so as to obtain one observation by stock. Second, we compute the mean of P50 across the 74 stocks of our sample. As an example, the mean of Total trades in panel A is the mean across stocks of the median number of trades within a 1.5 s interval. Note that Share volume represents round lots of 100 share units. Absolute log returns in Panel B are returns computed over the MFC interval and not over the 1.5 s interval. The table also reports univariate tests for means differences and covers the full sample of flash events with transaction proportions ranging from 0% to 100% on Nasdaq. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels

Net HFT-activity measures

We capture HFT- and NHFT-trading activity around price shocks via a measure of trade imbalance (a proxy for their respective behavior), following Brogaard et al. (2018). We compute trade imbalance for both HFTs and NHFTs as follows:

$$(N)HFT^D = (N)HFT^{D^+} - (N)HFT^{D^-} \quad (4.5)$$

$$(N)HFT^S = (N)HFT^{S^+} - (N)HFT^{S^-} \quad (4.6)$$

where $(N)HFT^D$ represents the liquidity demanded by (N)HFTs, $(N)HFT^S$ represents the liquidity supplied by (N)HFTs, $(N)HFT^{D^+}$ and $(N)HFT^{S^+}$ represent the liquidity demanded and supplied in the direction of the crash, respectively, and $(N)HFT^{D^-}$ and $(N)HFT^{S^-}$ represent the liquidity demanded and supplied in the opposite direction of the crash, respectively.

We then compute the net imbalance for both HFTs (HFT^{NET}) and NHFTs ($NHFT^{NET}$), where HFT^{NET} is the sum of HFT^D and HFT^S , and $NHFT^{NET}$ is the sum of $NHFT^D$ and $NHFT^S$. Net imbalance indicates the direction of net trading activity relative to a crash. A positive net imbalance implies trading activity moves in the same direction of the crash on an aggregate basis. This hints at “vicious” and destabilizing trading conduct. In contrast, a negative net imbalance implies trading activity moves in the opposite direction as the crash on an aggregate basis. This points to “virtuous” market-stabilizing activity.

Descriptive statistics

Table 1 presents the descriptive statistics of our MFC and SFE samples and Table 2 presents those of our OC-EPM and HL-EPM samples. For comparison, both tables include the full sample. We perform a hypothesis test for the difference in means to determine whether the trading-activity statistics (total trades, share volume, and dollar volume), liquidity statistics (quoted spread, relative spread, depth, and dollar depth), and proportion statistics (proportions of HFT trades, HFT shares, and HFT volume) are statistically different between the full sample and the MFC, SFE, OC-EPM, and HL-EPM samples.

Table 2 Summary statistics for the Full, OC-EPM, and HL-EPM samples

Panel A: Full sample					
	Minimum	Median	Mean	Maximum	Std Dev
Absolute log return, %	0	0	0.0119	21.86	0.0336
Total trades (all U.S. exchanges)	0	0	3.76	1,205	11.25
Total trades (Nasdaq)	0	0	1.57	880	5.45
Proportion HFT trades (Nasdaq)	0	0	0.1821	1	0.3611
Proportion HFT shares (Nasdaq)	0	0	0.1784	1	0.3591
Proportion HFT volume (Nasdaq)	0	0	0.1784	1	0.3591
Share volume	0	0	850.35	28,368,232	10,720.59
Dollar volume	0	0	36,099.82	1,458,100,000	459,793.2
Depth	2	6	23.77	100,816	100.00
Dollar Depth	6.34	152.75	365.57	2,225,817	1819.76
Quoted spread, \$	0.01	0.02	0.0412	113.91	0.1143
Relative spread, %	0.0022	0.0631	0.1225	56.21	0.7775
Panel D: OC-EPM sample					
	Minimum	Median	Mean	Maximum	Std Dev
Absolute log return, %	0.1099	0.3563	0.4241***	21.86	0.4032
Total trades (all U.S. exchanges)	0	13	32.45***	1,054	55.89
Total trades (Nasdaq)	0	5	14.71***	672	28.52
Proportion HFT trades (Nasdaq)	0	0.6667	0.5538***	1	0.3962
Proportion HFT shares (Nasdaq)	0	0.5861	0.5227***	1	0.4000
Proportion HFT volume (Nasdaq)	0	0.5861	0.5227***	1	0.4000
Share volume	0	2,073	11,095***	10,356,583	79,381
Dollar volume	0	66,836	450,955***	1,458,100,000	8,852,987
Depth	2	5.14	13.68***	1,823.8	45.76
Dollar depth	6.58	124.19	241.51***	42,745	800
Quoted spread, \$	0.01	0.0468	0.1501***	113.91	1.3609
Relative spread, %	0.0024	0.163	0.3691***	42.89	0.8135
Panel E: HL-EPM sample					
	Minimum	Median	Mean	Maximum	Std Dev
Absolute log return, %	0.1357	0.4334	0.5278***	26.51	0.5702
Total trades (all U.S. exchanges)	0	13	33.10***	1,054	57.43
Total trades (Nasdaq)	0	5	14.99***	672	29.28
Proportion HFT trades (Nasdaq)	0	0.6429	0.5383***	1	0.3958
Proportion HFT shares (Nasdaq)	0	0.5528	0.5067***	1	0.3987
Proportion HFT volume (Nasdaq)	0	0.5530	0.5067***	1	0.3987
Share volume	0	2,187	13,906***	28,368,232	201,933.1
Dollar volume	0	70,605	547,667***	1,458,100,000	9,996,160
Depth	2	4.88	12.86***	1,824	48.45
Dollar depth	6.58	120.21	238.30***	75,122	920.21
Quoted spread, \$	0.01	0.0631	0.1721***	113.91	1.38
Relative spread, %	0.0024	0.2125	0.4313***	42.89	0.8533

The table reports descriptive statistics for the full sample (Panel A), the sample of OC-EPMs (Panel D), which follows the methodology proposed by Brogaard et al. (2018) and the sample of HL-EPMs (Panel E), which is an EPM identification methodology alternative to Brogaard et al. (2018) we propose. N corresponds to the number of time intervals in the first

Table 2 (continued)

regression model. The lower the number of ultra-flash events, the lower N. We use a fixed 1.5 s interval for Panels A, D and E. All data are from Tickdata except Total trades and Proportion of HFT trades, HFT shares and HFT volume which are from Nasdaq. The mean of Absolute log return, Total trades, Depth, Dollar volume, Share volume, Quoted spread and Relative spread is computed in two steps. First, we compute the P50 bystock so as to obtain one observation by stock. Second, we compute the mean of P50 across the 74 stocks of our sample. As an example, the mean of Total trades in panel A is the mean across stocks of the median number of trades within a 1.5 s interval. Note that Share volume represents round lots of 100 share units. The table also reports univariate tests for means differences and covers the full sample of flash events with transaction proportions ranging from 0% to 100% on Nasdaq. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels

Descriptive statistics for MFCs and SFEs

Panel B in Table 1 reports the descriptive statistics for the sample of 419 MFCs, following the MFC-identification methodology used by Nanex (2010). As expected, the absolute log return, trading activity (as measured by total trades, share volume, and dollar volume), and spread (as measured by quoted and relative spreads) are substantially larger during MFCs than during the average 1.5 s interval in the full sample (Panel A). This means that most variables are affected by the occurrence of MFCs. The mean absolute MFC log return is 0.5577%, while the full sample mean absolute log return is only 0.0119%. As such, the mean absolute MFC log return is more than 46 times larger than the mean absolute full sample log return. Trading activity is also materially higher during MFCs. While approximately 4 trades are executed, on average, per 1.5 s in the full sample (Panel A), approximately 99 trades are executed, on average, across all U.S. exchanges per 1.5 s during MFCs, which means there are, on average, approximately 25 times more trades per 1.5 s interval during MFCs. Similarly, share volume (resp. dollar volume) is 65 times (resp. 64 times) higher during MFCs based on the mean, the quoted spread is 22 times higher during MFCs based on the mean, and the relative spread is 4 times higher during MFCs based on the mean when compared to the full sample.

Finally, the means of depth and dollar depth in the two populations are not statistically different. Available quantities, measured in shares and dollars, are not materially affected by the occurrence of MFCs, which arguably occur too quickly for market participants to modify them. Note that some MFCs occur in less than 1 ms (presented as 0 ms in the table) within our sample, which is consistent with the fact the fastest HFTs act within 5 ms, while other relatively fast traders act at speed levels of 50 ms to 150 ms (Scholtus et al. 2014).

Panel C in Table 1 reports the descriptive statistics for the sample of 5,491 SFEs following the alternative identification methodology based on the SFEs we propose. Similar to MFCs, the trading activity (as measured by total trades, share volume, and dollar volume) and spread (as measured by quoted and relative spreads) are considerably larger during SFEs than during the average 1.5 s interval of the full sample (Panel A). Again, most variables are affected by the occurrence of SFEs. The average absolute SFE log return is 0.63%, which is 53 times larger than the mean absolute full sample log return. Trading activity is also materially higher during SFEs. While approximately 4 trades are executed, on average, per 1.5 s within the full sample (Panel A), approximately 21 trades are executed, on average, across all U.S. exchanges per 1.5 s during SFEs. This means there are, on average, approximately 5 times more trades per 1.5 s interval during SFEs. Share volume (resp. dollar volume) is 10 times (resp. 11 times) higher during SFEs, on average, and the quoted spread is approximately 5 times higher during SFEs based on

the mean. Finally, similarly to MFCs, the means of depth and dollar depth in the two populations are not statistically different.

Descriptive statistics for OC-EPMs and HL-EPMs

Table 2 reports the summary statistics for the sample of 28,825 OC-EPMs in Panel D, following the EPM identification methodology used by Brogaard et al. (2018). Panel E includes the same statistics for the HL-EPM sample (Panel E), which is the alternative EPM identification methodology that we propose based on the 1.5 s interval high and low prices. The full sample is again indicated for ease of comparison (Panel A).

Similar to the MFC and SFE samples, the OC-EPM sample presents an absolute log return, trading activity (as measured by total trades, share volume, and dollar volume), and spread (as measured by quoted and relative spreads) that are materially (and highly statistically) larger during OC-EPMs than during the average 1.5 s interval of the full sample. The mean absolute OC-EPM log return is 0.4241%, while the full sample mean absolute return is 0.0119%. As such, the mean absolute OC-EPM log return is approximately 36 times larger than the mean absolute full-sample return.

Trading activity also appears to be materially higher during OC-EPMs. While approximately 4 trades are executed, on average, per 1.5 s within the full sample (Panel A), approximately 32 trades are executed, on average, per 1.5 s during OC-EPMs, which corresponds to approximately 9 times more trades per 1.5 s interval during OC-EPMs. Similarly, share volume (resp. dollar volume) is approximately 13 times (resp. 12 times) higher during OC-EPMs based on the mean. Moreover, the quoted spread is approximately 4 times higher, and the relative spread is more than 3 times higher during OC-EPMs, based on the mean, when compared to the full sample.

Panel E reports the descriptive statistics for the sample of 28,831 HL-EPMs, following the alternative EPM identification methodology we propose. The descriptive statistics of HL-EPMs are very similar to those of OC-EPMs. Thus, they do not require further comments. Note, however, that the mean absolute log return for HL-EPMs is higher than for OC-EPMs due to the fact that this methodology measures EPMs from high to low (resp. from low to high) instead of from open to close, as in the original methodology. The mean absolute HL-EPM log return is thus 0.5278%, while it is only 0.4241% for OC-EPMs.

Methodology

We conduct a three-step empirical analysis. First, we perform an event study to assess the liquidity dynamics of HFTs and NHFTs around flash events without controlling for any factors. We then conduct a contemporaneous multivariate regression analysis, zooming in on the occurrence of these flash events while controlling for additional drivers. Finally, we examine the variables that potentially lead to future crashes, using a logistic regression framework (logit). For ease of comparison, we first focus on MFCs and SFEs and then replicate the three-step empirical analysis on EPMs in a separate

section to compare all the different methodologies more easily (MFCs vs. SFEs vs. OC-EPMs vs. HL-EPMs).

While flash events in financial markets can be triggered by a combination of factors, such as algorithmic trading or market manipulation, which generate a sudden imbalance between supply and demand, we do not control for these trading-activity drivers for the following reasons.⁷ As in Brogaard et al. (2018), we cannot control for market manipulation at the HFT-firm level because Nasdaq data are aggregated. To the best of our knowledge, there is also no specific flag for NHFT algorithmic trades, preventing us from controlling for algorithmic traders' specific behaviors as opposed to those of pure HFT traders. In any case, analyzing HFT behavior during ultra-fast flash events is relevant, regardless of whether the trigger is an imbalance caused by NHFT algorithmic traders or some form of market manipulation. The reaction of HFTs and NHFTs during and after a crash is at least as important as what triggers the crash.

Empirical analysis

HFT activity around MFC and SFE events

Our initial approach involves an event study, designed to evaluate HFTs' and NHFTs' liquidity behavior during MFCs and SFEs. We aim to understand their respective trading activities by quantifying net directional trade imbalance, as conceptualized by Brogaard et al. (2018).⁸ A positive net imbalance ($HFT^{NET} > 0$) implies trading activity in the direction of a crash on an aggregate basis, which can destabilize the market and lead to liquidity black holes, whether observed during the pre-crash phase (crash triggering), the crash phase (crash exacerbation), or during the post-crash phase (resiliency prevention). In contrast, a negative net imbalance ($HFT^{NET} < 0$) implies trading activity in the opposite direction of a crash on an aggregate basis, which helps stabilize the market in the pre-crash phase (crash prevention), the crash phase (crash absorption), or during the post-crash phase (resiliency promotion).

We compute net imbalance for both types of traders (HFTs and NHFTs) at time intervals $t - 2$, $t - 1$, t , $t + 1$, and $t + 2$ for MFCs and SFEs, using fixed 1.5-second intervals. We focus, in particular, on time intervals $t - 1$, t , and $t + 1$, corresponding to the pre-crash, crash, and post-crash phases, respectively. Time intervals $t - 2$ and $t + 2$ are included in the analysis only to determine whether the pattern observed at $t - 1$ and $t + 1$ persists compared to the prior or successive interval.

We apply a filter to the MFC and SFE samples to prevent consecutive crashes from polluting the different windows from time interval $t - 2$ to time interval $t + 2$. For example, if a crash occurs at time interval t and another crash occurs at time interval $t + 1$, both crashes are removed from our sample to prevent any bias in the computation of the directional trade-imbalance metrics. Moreover, we remove overlapping crashes (across stocks) to prevent imbricated crashes from impacting our results. Our final samples comprise 419 MFCs and 5,491 SFEs after the cleaning process.

⁷ Technical glitches are filtered out in our SFE methodology because we eliminate the price-change rule and keep the most extreme sequences of consecutive tick movements with an absolute log return surpassing the 99.9th percentile of the absolute log return distribution. In addition, we focus on liquidity-driven flash events and filter out information-driven, news-related shocks because we require a 100% retracement in the second sequence in our SFE methodology.

⁸ Bellia et al. (2022) use a similar measure of net directional trade imbalance. They use monetary volume instead of share volume.

Table 3 (N)HFT Liquidity Supply and Demand around MFCs and SFEs

MFCs					
	$t - 2$	$t - 1$	t	$t + 1$	$t + 2$
		pre-crash	crash	post-crash	
HFT^{NET}	-24.0432	17.0460	-30.2291	13.3256	-34.0058
HFT^D	27.2536	47.5287**	298.3**	67.8501***	89.7291***
HFT^S	-51.2968	-30.4828	-328.5***	-54.5245***	-123.7***
$NHFT^{NET}$	24.0432	-17.0460	30.2291	-13.3256	34.0058
$NHFT^D$	132.0	248.6***	932.3*	149.7***	229.9**
$NHFT^S$	-107.9	-265.7***	-902.0*	-163.0***	-195.9**
SFEs					
	$t - 2$	$t - 1$	t	$t + 1$	$t + 2$
		pre-crash	crash	post-crash	
HFT^{NET}	-560.2***	-962.6***	-222.9***	-680.9***	-512.9***
HFT^D	429.6***	669.4***	218.6***	580.6***	407.3***
HFT^S	-989.8***	-1,632***	-441.5***	-1,261.5***	-920.2***
$NHFT^{NET}$	560.2***	962.6***	222.9***	680.9***	512.9***
$NHFT^D$	989.8***	1,632***	441.5***	1,261.5***	920.2***
$NHFT^S$	-429.6***	-669.4***	-218.6***	-580.6***	-407.3***

The table reports directional net trade imbalance around MFCs and SFEs. The metric is based on share volume and computed from Nasdaq. Time interval t is the sub-1.5-second interval corresponding to the crash. We also report the trade imbalance 3 s prior to the crash ($t - 2, t - 1$) and 3 s after the crash ($t + 1, t + 2$). HFT^{NET} (resp. $NHFT^{NET}$) is the sum of HFT^D and HFT^S (resp. $NHFT^D$ and $NHFT^S$) and $HFT^{NET} = -NHFT^{NET}$ and vice versa. When HFT^{NET} ($NHFT^{NET}$) is positive, it implies that HFTs (NHFTs) are trading in the direction of the return and thus in the direction of the crash (contributing to price instability) on an aggregate basis. When HFT^{NET} ($NHFT^{NET}$) is negative, it implies that HFTs (NHFTs) are trading in the opposite direction of the return and thus in the opposite direction of the crash (contributing to price stability) on an aggregate basis. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels

Table 3 presents the liquidity dynamics and net trading imbalance of HFTs and NHFTs around MFCs and SFEs.

MFCs

When using the MFC methodology, HFT^{NET} is not statistically significant in any time window, which implies that one cannot reject the null hypothesis that net imbalance from both HFTs and NHFTs is zero in the population. Thus, the MFC methodology does not provide much insight into the liquidity dynamics during crashes. Consequently, due to the lack of statistical significance, one cannot conclude who, among HFTs and NHFTs, triggers MFCs, exacerbates the crash, or enables stock prices to recover.

SFEs

When using the SFE methodology, HFT^{NET} becomes highly statistically significant in all time windows. This is due to both the ability to pinpoint the exact time of the crash while reconstructing the window around it and the larger SFE sample (5,491 crashes) compared to the MFC sample (419 crashes), both of which improve the overall significance level.

We focus, in turn, on the pre-crash, crash, and post-crash phases. Just before the crash, at time interval $t - 1$, we find that HFT^{NET} is negative (-962.6***) and statistically significant at the 1% level, which implies that HFTs, on an aggregate basis, trade in the opposite direction of the crash to help stabilize the market price. As such, they do not trigger the crash. During the crash, at time interval t , HFT^{NET} is again negative (-222.9***) and statistically significant at the 1% level, which implies that HFTs continue to trade in the opposite direction of the crash, contributing to price stability. Consequently, HFTs do not exacerbate the crash. In contrast, NHFTs trade in the direction of the crash. We also note that HFTs supply twice as much liquidity during the crash ($441.5^{***}/218.6^{***}=2$) in the opposite direction of the crash as NHFTs, while consuming only half the liquidity ($=218.6^{***}/441.5^{***}=0.5$) in the direction of the crash compared to NHFTs. Finally, directly after the crash, at time interval $t + 1$, HFT^{NET} is again negative (-680.9***) and statistically significant at the 1% level, which implies that HFTs continue trading in the opposite direction of the crash. Thus, they do contribute to the price recovery and market stability.

Based on this event study on SFEs, HFTs contribute to price stability across all event windows. As revealed in the next section, however, we find different results when conducting a multivariate regression analysis that accounts for all key trading-activity drivers present at the time of the flash event.

HFT activity and other key trading drivers during MFC and SFE events

An important aspect of our methodological approach must be emphasized here. Instead of considering all flash events occurring on all U.S. exchanges in the following regressions, we only consider flash events in which the proportion of transactions occurring on Nasdaq during the interval represents at least 50% of all transactions on U.S. stock exchanges. Transactions may occur on different exchanges during flash events, and they sometimes occur outside the Nasdaq exchange. Thus, we filter out flash events that are not prevalent on Nasdaq, which enables us to focus on HFTs' activity on Nasdaq (for which we possess information via a flag) during crashes that partially or totally occur on Nasdaq. We use an event window of 15 s (10 intervals) before and after the event in our base case methodology.⁹

Following Brogaard et al. (2018), we use the following multivariate model specification:

$$HFT_{it}^{NET} = \alpha_0 + \beta_1 EVENT_{it} + \beta_2 Abs Ret_{it} + \beta_3 SV_{it} + \beta_4 RS_{it} + Lags_{kit-\sigma} \Gamma_{k\sigma} + \varepsilon_{it} \quad (5.1)$$

where HFT_{it}^{NET} is the difference between HFT^D and HFT^S , $EVENT_{it}$ is a dummy variable equal to one if the 1.5-second interval t in stock i corresponds to an MFC or an SFE and zero otherwise; $AbsRet_{it}$ is the absolute log return, SV is the share volume, RS is the relative spread, and the vector, $\Gamma_{k\sigma}$, includes both the dependent variable and all of the independent variables of the regression, which are indexed with a subscript k and lag $\sigma \in$

⁹ As mentioned in Section 5.5, we perform several robustness checks. First, we conduct the regressions on the full flash-event sample, i.e., flash events where the proportion of transactions on Nasdaq is between 0% and 100%. We also use 6-second (4 intervals) and 9-second (6 intervals) windows before and after the event.

Table 4 Multivariate regressions - Net HFT imbalance during crashes

Proportion of transactions on Nasdaq $\geq 50\%$			
MFCs			
	(5.1)	(5.2)	(5.3)
<i>MFC</i>	-1.1459**		-0.9314
<i>MFC^{STANDALONE}</i>		-1.4241**	
<i>MFC^{SIMULTANEOUS}</i>		-0.6728	
<i>EXTREME_HOURS</i>			0.0656
<i>MFC * EXTREME_HOURS</i>			-0.6015
<i>AbsRet</i>	-0.0097	-0.0102	-0.0091
<i>SV</i>	-0.0051	-0.0043	-0.0056
<i>RS</i>	0.0333	0.0341	0.0384
<i>Adj.R²</i>	0.009	0.012	0.009
<i>N</i>	7,072	7,072	7,072
Proportion of transactions on Nasdaq $\geq 50\%$			
SFEs			
	(5.1)	(5.2)	(5.3)
<i>SFE</i>	0.1762***		0.1841***
<i>SFE^{STANDALONE}</i>		0.1762***	
<i>SFE^{SIMULTANEOUS}</i>		0.1692***	
<i>EXTREME_HOURS</i>			-0.0199
<i>SFE * EXTREME_HOURS</i>			-0.0430
<i>AbsRet</i>	-0.0067	-0.0067	-0.0067**
<i>SV</i>	-0.0188	-0.0186	-0.0185
<i>RS</i>	-0.0119**	-0.0120**	-0.0118**
<i>Adj.R²</i>	0.052	0.052	0.052
<i>N</i>	69,781	69,781	69,781

The table reports the estimated coefficients of equations 5.1, 5.2 and 5.3 where HFT^{NET} is the difference between HFT^D , the liquidity demanded by HFTs, and HFT^S , the liquidity supplied by HFTs. A negative HFT^{NET} implies an aggregate trading activity in the opposite direction of the crash (contributing to price stability) while a positive HFT^{NET} implies an aggregate trading activity in the direction of the crash (contributing to price instability). MFC_{it} (resp. SFE_{it}) is a dummy variable equal to one if the 1.5-second interval t in stock i is identified as an MFC (resp. SFE) and is equal to zero otherwise. $MFC^{STANDALONE}$ (resp. $SFE^{STANDALONE}$) is a dummy that captures MFCs (resp. SFEs) that occur on their own, $MFC^{SIMULTANEOUS}$ (resp. $SFE^{SIMULTANEOUS}$) is a dummy that captures MFCs (resp. SFEs) that occur on several sample stocks within the same time interval, $EXTREME_HOURS$ is a dummy that captures transactions identified as occurring during extreme hours (EH), i.e., from 9:00 a.m. to 9:05 a.m. and from 15:55 p.m. to 16:00 p.m., $MFC * EXTREME_HOURS$ (resp. $SFE * EXTREME_HOURS$) is an interaction variable that captures the additional effect of MFCs (resp. SFEs) that occur in the first five and last five minutes of the trading day, $AbsRet$ is the absolute log return, SV is the share volume and RS is the relative spread. The regressions are run on an MFC (resp. SFE) subsample whose proportion of transactions on Nasdaq is set to at least 0.5, meaning that at least 50% of transactions during the crash occur on Nasdaq. All the non-dummy variables are standardized at the stock level and regressions are run with stock-fixed effects. The number of observations (N) is not equal to the number of shocks but to the total number of observations during the shocks as well as the lags. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels

$\{1,2,...,10\}$. Thus, $Lags_{kit-\sigma}$ is a vector of 10 lags. All the non-dummy variables are standardized at the stock level and we include stock-fixed effects.

We also estimate two alternative specifications of the model, allowing us to conduct the multivariate regression by focusing, in turn, on all flash events, standalone and simultaneous flash events, as well as flash events occurring exclusively during extreme hours. Standalone events occur in isolation, while simultaneous events occur across

multiple stocks within the same time interval. The first alternative specification is as follows:

$$HFT_{it}^{NET} = \alpha_0 + \beta_1 EVENT_{it}^{STA} + \beta_2 EVENT_{it}^{SIM} + \beta_3 AbsRet_{it} + \beta_4 SV_{it} + \beta_5 RS_{it} + \mathbf{Lags}_{kit-\sigma} \mathbf{\Gamma}_{k\sigma} + \epsilon_{it} \quad (5.2)$$

where all the variables are as previously defined, $EVENT_{it}^{STA}$ is a dummy variable equal to one if the 1.5-second interval t in stock i is identified as a standalone flash event and is equal to zero otherwise; $EVENT_{it}^{SIM}$ is a dummy variable equal to one if the 1.5-second interval t in stock i is identified as a simultaneous flash event and is equal to zero otherwise. The second alternative specification is as follows:

$$HFT_{it}^{NET} = \alpha_0 + \beta_1 EVENT_{it} + \beta_2 EH_{it} + \beta_3 EVENT_{it} * EH_{it} + \beta_4 AbsRet_{it} + \beta_5 SV_{it} + \beta_6 RS_{it} + \mathbf{Lags}_{kit-\sigma} \mathbf{\Gamma}_{k\sigma} + \epsilon_{it} \quad (5.3)$$

where all the variables are as previously defined, EH_{it} is a dummy variable equal to one if the 1.5-second interval t in stock i is identified as occurring during extreme hours (EH), i.e., during the first five and last five minutes of the trading day, and $EVENT_{it} * EH_{it}$ is an interaction variable that captures the additional effect of flash events that occur during extreme hours (EH).

The results of the different multivariate regressions for both MFCs and SFEs are presented in Table 4.

The SFE methodology yields the most enlightening and consistent outcomes. When pinpointing the exact time of a crash and rebuilding the window around the crash, which is what we do with our SFE methodology, we find that the coefficient of the SFE dummy variable is always positive and statistically significant at the 1% level in all specifications, implying that HFTs increase their liquidity demand, on aggregate, during the extreme price shock and trade in the direction of the crash when it occurs. This is not specific to extreme hours, for which we find no statistical significance. We turn to the control variables and find that the relative spread is the only statistically significant variable at the 5% level in all SFE regressions. This indicates that HFTs provide more liquidity than they absorb when spreads widen in normal market conditions, i.e., in the absence of shocks. Overall, HFTs seem to play a positive role in normal times and a negative one at the time of the shocks. Regarding MFCs, we find much weaker evidence. The total number of observations is also 10 times lower than in the SFE specification (considering all observations during the shocks as well as the lags).

Predicting flash events based on prior HFT activity

In this subsection, we examine the variables that potentially lead to the occurrence of future flash events, whether these events are MFCs or SFEs. We use the logistic regression framework to determine what drives future shocks, with the dependent variable being a binary response variable. Our objective is to examine whether HFTs' activity at time interval $t - 1$ explains the flash event at time interval t . Our model is specified as follows:

$$\begin{aligned} Prob(EVENT = 1)_{it} = & \alpha_0 + \beta_1 HFT^{NET}_{it-1} + \beta_2 AbsRet_{it-1} \\ & + \beta_3 SV_{it-1} + \beta_4 RS_{it-1} + \beta_5 HFTP_{it-1} + \epsilon_{it} \end{aligned} \quad (5.4)$$

where *EVENT* is a binary variable equal to one if the 1.5-second interval *t* contains an MFC (resp. an SFE) on stock *i* and zero otherwise, HFT^{NET} corresponds to the difference between HFT^D and HFT^S as defined above, *AbsRet* is the absolute log return, *SV* is the share volume, and *RS* is the relative spread. We follow Brogaard et al. (2018) but add a supplementary explanatory variable, i.e., HFT participation based on trades (*HFTP*), to complement the initial model, following Carrion (2013) and Leal et al. (2016). All non-dummy variables are standardized at the stock level and lagged by one period. We again consider MFCs and SFEs for which the proportion of transactions occurring on Nasdaq during the crash represents at least 50% of all transactions on U.S. exchanges, thus filtering out MFCs and SFEs that are not prevalent on Nasdaq.

It is important to note that a logit model may be biased when events and non-events are imbalanced, which is the case in our model, as MFCs, in particular, are rare events. Thus, due to the potential bias engendered by the imbalance between events and non-events within our samples (the number of MFCs and SFEs vs. the number of non-MFCs and non-SFEs), we implement a penalized maximum likelihood estimation, as advocated by Firth (1993).

We also estimate three alternative specifications of the logit model. Instead of considering all shocks, as in Equation 5.4, we focus on standalone shocks (*STAEVENT*) in Equation 5.5, simultaneous shocks (*SIMEVENT*) in Equation 5.6, and shocks occurring exclusively during extreme hours (*EHEVENT*) in Equation 5.7. The three alternative specifications are as follows:

$$\begin{aligned} Prob(STAEVENT = 1)_{it} = & \alpha_0 + \beta_1 HFT^{NET}_{it-1} + \beta_2 AbsRet_{it-1} \\ & + \beta_3 SV_{it-1} + \beta_4 RS_{it-1} + \beta_5 HFTP_{it-1} + \epsilon_{it} \end{aligned} \quad (5.5)$$

$$\begin{aligned} Prob(SIMEVENT = 1)_{it} = & \alpha_0 + \beta_1 HFT^{NET}_{it-1} + \beta_2 AbsRet_{it-1} \\ & + \beta_3 SV_{it-1} + \beta_4 RS_{it-1} + \beta_5 HFTP_{it-1} + \epsilon_{it} \end{aligned} \quad (5.6)$$

$$\begin{aligned} Prob(EHEVENT = 1)_{it} = & \alpha_0 + \beta_1 HFT^{NET}_{it-1} + \beta_2 AbsRet_{it-1} \\ & + \beta_3 SV_{it-1} + \beta_4 RS_{it-1} + \beta_5 HFTP_{it-1} + \epsilon_{it} \end{aligned} \quad (5.7)$$

The results for the MFCs are reported in Table 5. The main finding is the lack of overall statistical significance of our variable of interest. HFT^{NET} is not statistically significant in any specification of the logit model. Consequently, we cannot reject the null hypothesis that HFT-trading activity before an extreme price shock plays no role in explaining the shock. The two most significant control variables are the absolute log return (*AbsRet*) and relative spread (*RS*). They both display positive and statistically significant coefficients.

When we consider all shocks, we estimate that a one-standard deviation increase in the absolute log return raises the odds of having an MFC in the following interval by 19.2%. This effect is even stronger when focusing only on simultaneous shocks, where the odds increase to 33.3%. The relative spread also plays a pivotal role as a one-standard

Table 5 LOGIT - MFCs

Proportion of transactions on Nasdaq $\geq 50\%$					
	Coefficient	Odds ratio	P-value	C-stat	N
All (Nb of MFCs=114)				0.554	2,067
α_0	-4.1755		***		
HFT_{it-1}^{NET}	-0.0182	0.982			
$AbsRet_{it-1}$	0.1755	1.192	***		
SV_{it-1}	0.0257	1.026			
RS_{it-1}	0.0901	1.094			
$HFTP_{it-1}$	-0.1300	0.878			
Standalone (Nb of MFCs=77)				0.545	1,396
α_0	-4,5102		***		
HFT_{it-1}^{NET}	-0.0906	0.913			
$AbsRet_{it-1}$	0.0470	1.048			
SV_{it-1}	0.0475	1.049			
RS_{it-1}	-0.0666	0.936			
$HFTP_{it-1}$	-0.0241	0.976			
Simultaneous (Nb of MFCs=37)				0.665	671
α_0	-5,4716		***		
HFT_{it-1}^{NET}	0.0896	1.094			
$AbsRet_{it-1}$	0.2874	1.333	***		
SV_{it-1}	0.0431	1.044			
RS_{it-1}	0.2200	1.246	***		
$HFTP_{it-1}$	-0.3201	0.726			
Extreme hours (Nb of MFCs=41)				0.658	743
α_0	-5,2805		***		
HFT_{it-1}^{NET}	-0.0136	0.986			
$AbsRet_{it-1}$	0.1473	1.159			
SV_{it-1}	0.0485	1.050			
RS_{it-1}	0.2860	1.331	***		
$HFTP_{it-1}$	-0.2232	0.800			

The table reports the results of equations 5.4, 5.5, 5.6, and 5.7 regarding the probability for a stock to undergo an MFC at time interval t as a function of HFT_{it-1}^{NET} , absolute log return ($AbsRet$), share volume (SV), relative spread (RS) and HFT participation based on trades ($HFTP$) at time interval $t - 1$. The event window is 15 s before and after the crash. All the non-dummy variables are standardized at the stock level and lagged by one period except the intercept. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels. The c-statistic measures the goodness of fit for binary outcomes in the logistic regression model. As an illustration, if we randomly choose two stocks from our sample, one that has the outcome characteristic (experienced an MFC) and one that does not have the outcome characteristic (did not experience an MFC), each stock has a predicted probability of experiencing a crash from the logistic regression model. The c-statistic corresponds to the probability that the stock that truly has the outcome characteristic will have a higher predicted probability from the logistic regression equation than the stock that truly does not have the characteristic

deviation increase in the spread raises the odds of experiencing a simultaneous MFC in the following interval by 24.6%. The increase in the odds is even higher (33.1%) when the shock occurs during extreme hours.

The results for the SFEs are reported in Table 6. We find that our SFE methodology greatly improves the overall statistical significance of the model when compared to the MFC methodology. In our base model including all SFEs, the variable of interest, HFT_{it-1}^{NET} , displays a negative sign and is significant at the 1% level. A one-standard deviation increase in HFT_{it-1}^{NET} is associated with a decrease in the odds of having an SFE in the next interval by 4%. Therefore, we can reject the hypothesis that extreme price

Table 6 LOGIT - SFEs

Proportion of transactions on Nasdaq $\geq 50\%$					
	Coefficient	Odds ratio	P-value	C-stat	N
All (Nb of SFEs=5,491)				0.569	69,854
α_0	-2.4846		***		
HFT_{it-1}^{NET}	-0.0409	0.96	***		
$AbsRet_{it-1}$	0.0277	1.028	**		
SV_{it-1}	0.1648	1.179	***		
RS_{it-1}	-0.2039	0.816	***		
$HFTP_{it-1}$	-0.1157	0.891	***		
<i>Standalone (Nb of SFEs=2,047)</i>				0.581	26,041
α_0	-3.5319		***		
HFT_{it-1}^{NET}	-0.1277	0.88	***		
$AbsRet_{it-1}$	0.0253	1.026			
SV_{it-1}	0.1037	1.109	***		
RS_{it-1}	-0.2431	0.784	***		
$HFTP_{it-1}$	-0.0501	0.951	**		
<i>Simultaneous (Nb of SFEs=3,444)</i>				0.553	43,813
α_0	-2.9797		***		
HFT_{it-1}^{NET}	0.0261	1.026			
$AbsRet_{it-1}$	0.0293	1.03	*		
SV_{it-1}	0.0829	1.086	***		
RS_{it-1}	-0.1623	0.85	***		
$HFTP_{it-1}$	-0.1419	0.868	***		
<i>Extreme hours (Nb of SFEs=1,362)</i>				0.629	10,067
α_0	-3.9974		***		
HFT_{it-1}^{NET}	-0.0157	0.984			
$AbsRet_{it-1}$	0.0934	1.098	***		
SV_{it-1}	0.1549	1.168	***		
RS_{it-1}	0.0664	1.069	***		
$HFTP_{it-1}$	-0.3386	0.713	***		

The table reports the results of equations 5.4, 5.5, 5.6, and 5.7 regarding the probability for a stock to undergo an SFE at time interval t as a function of HFT_{it-1}^{NET} , absolute log return ($AbsRet$), share volume (SV), relative spread (RS) and HFT participation based on trades ($HFTP$) at time interval $t-1$. The event window is 15 s before and after the crash. All the non-dummy variables are standardized at the stock level and lagged by one period except the intercept. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels. The c-statistic measures the goodness of fit for binary outcomes in the logistic regression model. As an illustration, if we randomly choose two stocks from our sample, one that has the outcome characteristic (experienced an SFE) and one that does not have the outcome characteristic (did not experience an SFE), each stock has a predicted probability of experiencing a crash from the logistic regression model. The c-statistic corresponds to the probability that the stock that truly has the outcome characteristic will have a higher predicted probability from the logistic regression equation than the stock that truly does not have the characteristic

shocks are related to HFTs consuming more liquidity than they provide during the preceding time interval. This is particularly true for standalone shocks, for which the odds decrease by 12%. The level of HFT participation in the market ($HFTP$) is also negatively associated with the occurrence of a flash event during the next period. All else being equal, a one-standard deviation increase in $HFTP$ leads to a decrease in the odds of having an SFE in the next interval by 10.9%. This stabilizing dynamic is observed in all four specifications, with the most pronounced stabilizing effect observed during shocks that occur in extreme hours, where the odds decrease by nearly 30%.

Table 7 (N)HFT Liquidity Supply and Demand around OC-EPMs and HL-EPMs

OC-EPMs					
	$t - 2$	$t - 1$	t	$t + 1$	$t + 2$
		pre-crash	crash	post-crash	
HFT^{NET}	3.4496**	0.9200	15.4702***	10.0552***	3.3120**
HFT^D	-0.0190	13.4265***	139.2***	5.5128***	-4.4022***
HFT^S	3.4686***	-12.5065***	-123.7***	4.5424***	7.7142***
$NHFT^{NET}$	-3.4496**	-0.9200	-15.4702***	-10.0552***	-3.3120**
$NHFT^D$	3.0694	38.0065***	291.7***	32.8409***	10.7208***
$NHFT^S$	-6.5190	-38.9265***	-307.2***	-42.8961***	-14.0328***
HL-EPMs					
	$t - 2$	$t - 1$	t	$t + 1$	$t + 2$
		pre-crash	crash	post-crash	
HFT^{NET}	2.3880	2.8856	8.7507	12.9138***	2.1112
HFT^D	-0.3184	11.5792***	118.8***	5.1912***	-4.4756***
HFT^S	2.7064	-8.6936***	-110.1***	7.7225***	6.5868***
$NHFT^{NET}$	-2.3880	-2.8856	-8.7507	-12.9138***	-2.1112
$NHFT^D$	-2.8653	25.3466***	274.1***	22.5343***	7.6329***
$NHFT^S$	0.4773	-28.2322***	-282.8***	-35.4481***	-9.7441***

The table reports directional net trade imbalance around OC-EPMs and HL-EPMs. The metric is based on share volume and computed from Nasdaq. Time interval t is the sub-1.5-second interval corresponding to the crash. We also report the trade imbalance 3 s prior to the crash ($t - 2, t - 1$) and 3 s after the crash ($t + 1, t + 2$). HFT^{NET} (resp. $NHFT^{NET}$) is the sum of HFT^D and HFT^S (resp. $NHFT^D$ and $NHFT^S$) and $HFT^{NET} = -NHFT^{NET}$ and vice versa. When HFT^{NET} ($NHFT^{NET}$) is positive, it implies that HFTs (NHFTs) are trading in the direction of the return and thus in the direction of the crash (contributing to price stability) on an aggregate basis. When HFT^{NET} ($NHFT^{NET}$) is negative, it implies that HFTs (NHFTs) are trading in the opposite direction of the return and thus in the opposite direction of the crash (contributing to price stability) on an aggregate basis. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels

The control variable that provides the most valuable insight in Table 6 is the share volume (SV). Its coefficient consistently exhibits a positive and statistically significant relationship. Its economic significance is particularly pronounced when considering all SFEs. A one-standard deviation increase in the share volume increases the odds of having an SFE in the following interval by 17.9%. This variable is the most critical driver contributing to the occurrence of a flash event during the next period.

Comparison with the so-called EPMs

In this section, we provide a parallel analysis based on EPMs. After comparing our sample of SFEs with MFCs, as identified by the Nanex detection algorithm, we compare our sample of SFEs with EPMs using close and open prices (Brogaard et al. 2018), as well as high and low prices in an alternative setting that we propose to use.

Table 8 Multivariate regressions - Net HFT Activity during crashes

Proportion of transactions on Nasdaq $\geq 50\%$			
OC-EPMs			
	(5.1)	(5.2)	(5.3)
EPM^{OC}	15.3355		32.3419**
$EPM^{OC-STANDALONE}$		33.8257	
$EPM^{OC-SIMULTANEOUS}$		7.6939	
$EXTREME_HOURS$			-19.5770**
$EPM^{OC} * EXTREME_HOURS$			-88.7921***
$AbsRet$	0.9956	1.1002	1.7089
SV	-9.5890**	-9.8966**	-9.4521**
RS	-10.6984**	-10.5563**	-9.6340**
$Adj.R^2$	0.0113	0.0114	0.0121
N	136,221	136,221	136,221
Proportion of transactions on Nasdaq $\geq 50\%$			
HL-EPMs			
	(5.1)	(5.2)	(5.3)
EPM^{HL}	5.3031		16.8067
$EPM^{HL-STANDALONE}$		30.1423	
$EPM^{HL-SIMULTANEOUS}$		-4.6606	
$EXTREME_HOURS$			-23.2225***
$EPM^{HL} * EXTREME_HOURS$			-46.6136
$AbsRet$	-3.3997	-3.2856	-2.9449
SV	-6.3452	-6.9121	-6.3069
RS	-5.5217	-5.3650	-4.5542
$Adj.R^2$	0.0109	0.0111	0.0113
N	114,289	114,289	114,289

The table reports the estimated coefficients of equations 5.1, 5.2 and 5.3 where HFT^{NET} is the difference between HFT^D , the liquidity demanded by HFTs, and HFT^S , the liquidity supplied by HFTs. A negative HFT^{NET} implies an aggregate trading activity in the opposite direction of the crash (contributing to price stability) while a positive HFT^{NET} implies an aggregate trading activity in the direction of the crash (contributing to price instability). EPM^{OC} (resp. EPM^{HL}) is a dummy variable equal to one if the 1.5-second interval t in stock i is identified as an OC-EPM (resp. HL-EPM) and is equal to zero otherwise. $EPM^{OC-STANDALONE}$ (resp. $EPM^{HL-STANDALONE}$) is a dummy that captures OC-EPMs (resp. HL-EPMs) that occur on their own, $EPM^{OC-SIMULTANEOUS}$ (resp. $EPM^{HL-SIMULTANEOUS}$) is a dummy that captures OC-EPMs (resp. HL-EPMs) that occur on several sample stocks within the same time interval, $EXTREME_HOURS$ is a dummy that captures transactions identified as occurring during extreme hours (EH), i.e., from 9:00 a.m. to 9:05 a.m. and from 15:55 p.m. to 16:00 p.m., $EPM^{OC} * EXTREME_HOURS$ (resp. $EPM^{HL} * EXTREME_HOURS$) is an interaction variable that captures the additional effect of OC-EPMs (resp. HL-EPMs) that occur in the first five and last five minutes of the trading day, $AbsRet$ is the absolute log return, SV is the share volume and RS is the relative spread. The regressions are run on an OC-EPM (resp. HL-EPM) subsample whose proportion of transactions on Nasdaq is set to at least 0.5, meaning that at least 50% of transactions during the crash occur on Nasdaq. All the non-dummy variables are standardized at the stock level and regressions are run with stock fixed effects. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels

HFT activity around OC-EPMs and HL-EPMs

Table 7 presents the trading dynamics of HFTs and NHFTs around events characterized by OC-EPMs and HL-EPMs. Table 7 is similar to Table 3. As with MFCs and SFEs, we apply the same filters to the OC-EPM and HL-EPM samples to prevent consecutive crashes from impacting the different windows from time interval $t - 2$ to time interval $t + 2$. Our final samples comprise 28,825 OC-EPMs and 24,831 HL-EPMs after the cleaning.

Table 9 LOGIT - OC-EPMs

Proportion of transactions on Nasdaq $\geq 50\%$					
	Coefficient	Odds ratio	P-value	C-stat	N
All (Nb of OC-EPMs=7,785)				0.540	138,624
α_0	-2.8883		***		
HFT_{it-1}^{NET}	-0.0001	1.000	**		
$AbsRet_{it-1}$	0.0672	1.070	***		
SV_{it-1}	0.0568	1.058	***		
RS_{it-1}	0.0367	1.037	***		
$HFTP_{it-1}$	0.2049	1.227	***		
<i>Standalone (Nb of OC-EPMs=2,245)</i>				0.634	39,973
α_0	-3.9544		***		
HFT_{it-1}^{NET}	-0.0001	1.000	**		
$AbsRet_{it-1}$	0.0661	1.068	***		
SV_{it-1}	0.0869	1.091	***		
RS_{it-1}	-0.3846	0.681	***		
$HFTP_{it-1}$	-0.6447	0.525	***		
<i>Simultaneous (Nb of OC-EPMs=5,540)</i>				0.579	98,651
α_0	-3.3729		***		
HFT_{it-1}^{NET}	-0.0001	1.000	***		
$AbsRet_{it-1}$	0.0681	1.070	***		
SV_{it-1}	0.0337	1.034	***		
RS_{it-1}	0.1036	1.109	***		
$HFTP_{it-1}$	0.5101	1.665	***		
<i>Extreme hours (Nb of OC-EPMs=1,680)</i>				0.677	29,610
α_0	-4.5539		***		
HFT_{it-1}^{NET}	-0.0001	1.000	***		
$AbsRet_{it-1}$	0.1023	1.108	***		
SV_{it-1}	0.0480	1.049	***		
RS_{it-1}	0.2831	1.327	***		
$HFTP_{it-1}$	0.1709	1.186	***		

Table 9 reports the results of equations 5.4, 5.5, 5.6, and 5.7 regarding the probability for a stock to undergo an OC-EPM at time interval t as a function of HFT_{it-1}^{NET} , absolute log return ($AbsRet$), share volume (SV), relative spread (RS) and HFT participation based on trades ($HFTP$) at time interval $t - 1$. The event window is 15 s before and after the crash. All the non-dummy variables are standardized at the stock level and lagged by one period except the intercept. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels. The c-statistic measures the goodness of fit for binary outcomes in the logistic regression model. As an illustration, if we randomly choose two stocks from our sample, one that has the outcome characteristic (experienced an OC-EPM) and one that does not have the outcome characteristic (did not experience an OC-EPM), each stock has a predicted probability of experiencing a crash from the logistic regression model. The c-statistic corresponds to the probability that the stock that truly has the outcome characteristic will have a higher predicted probability from the logistic regression equation than the stock that truly does not have the characteristic

When using the OC-EPM methodology, HFT_{it-1}^{NET} is statistically significant in four out of five time windows. There is much weaker evidence when detecting EPMs based on the highs and lows. The two methods, nevertheless, agree on the role played by HFTs just before and after the shock. We find no evidence that HFTs trigger EPMs because their market activity is not statistically different from that observed for NHFTs before the crash. They, however, play a significant role after the crash. The coefficients of HFT_{it-1}^{NET} are positive (10.0552 and 12.9138, respectively) and statistically significant at the 1% level. When we consider the decomposition of HFT_{it-1}^{NET} , we observe that HFTs both supply and demand more liquidity

Table 10 LOGIT - HL-EPMs

Proportion of transactions on Nasdaq $\geq 50\%$					
	Coefficient	Odds ratio	P-value	C-stat	N
All (Nb of HL-EPMs=7,152)				0.575	117,126
α_0	-2.8357		***		
HFT_{it-1}^{NET}	-0.00004	1.000			
$AbsRet_{it-1}$	0.1589	1.172	***		
SV_{it-1}	0.0476	1.049	***		
RS_{it-1}	0.0521	1.053	***		
$HFTP_{it-1}$	0.2716	1.312	***		
<i>Standalone (Nb of HL-EPMs=2,085)</i>				0.628	34,145
α_0	-3.8669		***		
HFT_{it-1}^{NET}	-0.0001	1.000	**		
$AbsRet_{it-1}$	0.1406	1.151	***		
SV_{it-1}	0.0823	1.086	***		
RS_{it-1}	-0.3470	0.707	***		
$HFTP_{it-1}$	-0.5791	0.560	***		
<i>Simultaneous (Nb of HL-EPMs=5,067)</i>				0.614	82,961
α_0	-3.3408		***		
HFT_{it-1}^{NET}	-0.0001	1.000	**		
$AbsRet_{it-1}$	0.1556	1.168	***		
SV_{it-1}	0.0211	1.021	***		
RS_{it-1}	0.1240	1.132	***		
$HFTP_{it-1}$	0.5902	1.804	***		
<i>Extreme hours (Nb of HL-EPMs=1,945)</i>				0.704	31,853
α_0	-4.2947		***		
HFT_{it-1}^{NET}	-0.0001	1.000	**		
$AbsRet_{it-1}$	0.1616	1.175	***		
SV_{it-1}	0.0455	1.047	***		
RS_{it-1}	0.3105	1.364	***		
$HFTP_{it-1}$	0.2612	1.298	***		

Table 10 reports the results of equations 5.4, 5.5, 5.6, and 5.7 regarding the probability for a stock to undergo an HL-EPM at time interval t as a function of HFT_{it-1}^{NET} , absolute log return ($AbsRet$), share volume (SV), relative spread (RS) and HFT participation based on trades ($HFTP$) at time interval $t - 1$. The event window is 15 s before and after the crash. All the non-dummy variables are standardized at the stock level and lagged by one period except the intercept. Asterisks *** and ** indicate statistical significance at the 1% and 5% levels. The c-statistic measures the goodness of fit for binary outcomes in the logistic regression model. As an illustration, if we randomly choose two stocks from our sample, one that has the outcome characteristic (experienced an HL-EPM) and one that does not have the outcome characteristic (did not experience an HL-EPM), each stock has a predicted probability of experiencing a crash from the logistic regression model. The c-statistic corresponds to the probability that the stock that truly has the outcome characteristic will have a higher predicted probability from the logistic regression equation than the stock that truly does not have the characteristic

in the direction of the shock after it occurs in the market during the preceding interval. As opposed to NHFTs, HFTs do not contribute to price recovery.

HFT activity and other key trading drivers during EPM events

In this subsection, we focus on HFTs' trading activity during EPMs and control for all the key trading drivers. We perform the same multivariate regressions as in Section 5.2. The results of the different regressions are presented in Table 8.

We find very little evidence that flash events are related to HFTs' trading activity. The only case (out of six) in which HFTs' net trading activity is destabilizing

(because it follows the same direction as the shock) is in the model specification with OC-EPMs and extreme hours. However, there is no observed destabilizing activity for OC-EPMs that occur during extreme hours. In contrast, the interaction variable ($EPM^{OC} * EXTREME_HOURS$) is significant, and the size and negative sign of its coefficient indicate that HFTs help stabilize the market during price shocks that occur specifically during extreme hours. Except for this special case, there is overall insufficient statistical evidence to suggest that HFT activity varies significantly when EPMs occur in the market. Finally, we note that the overall statistical significance of the models is lower when the EPM methodology is used instead of ours, based on SFEs.

Predicting flash events based on prior HFT activity

In this subsection, we examine the variables that potentially lead to the occurrence of future crashes, whether these crashes are OC-EPMs or HL-EPMs, and we compare our results to those we obtain for MFCs and SFEs in Section 5.3. The results for the OC-EPMs and HL-EPMs are reported in Tables 9 and 10.

The results are very similar irrespective of the choice of open and close prices or high and low prices. Our variable of interest, HFT^{NET} , which measures the level of HFT-trading activity in the direction of the EPM, is significant in all model specifications, except in one (on all HL-EPMs). However, the economic significance of HFT-trading activity remains extremely limited. The odds of having an EPM in the following interval are practically unchanged across all model specifications. In fact, the level of HFT participation based on trades ($HFTP$) seems to matter more to anticipate the occurrence of an EPM. In all but one model specification, it displays a positive and significant coefficient estimate. A one-standard deviation increase in the level of HFT participation in the market based on trades raises the odds of having an EPM in the following interval from 18.6% for an OC-EPM occurring during extreme hours to 80.4% for a simultaneous HL-EPM.

The control variables are always statistically significant at the 1% level in both OC-EPM and HL-EPM model specifications, as indicated in Tables 9 and 10. Overall, the absolute log return and share volume variables are the most consistent drivers of EPMs (besides HFT participation).

Robustness checks

In the preceding subsections, we conducted a parallel analysis of three types of market events: MFCs, EPMs (both OC-EPMs and HL-EPMs), and SFEs, specifically examining their similarities and differences. Thus, we investigated whether our results were similar across methodologies or, conversely, methodology-specific. We found different results depending on the price-shock identification methodology used, which implies that concluding on the virtuousness or viciousness of HFTs may be primarily methodology-dependent. We also performed additional analyses using alternative time intervals of 1 second and 2 s, in addition to our base time interval of 1.5 s. Regardless of the time interval considered, our results remained highly consistent.

Our study focused on ultra-fast flash events (MFCs, EPMs, and SFEs), a subcategory of flash events, where the proportion of transactions occurring on Nasdaq during the

interval constituted at least 50% of all transactions on U.S. stock exchanges, thus filtering out price shocks that were not prevalent on Nasdaq. However, as a robustness check, we also conducted regressions on the full sample, which encompassed flash events with transaction proportions ranging from 0% to 100% on Nasdaq. No significant change was found in the results. This is due to the fact Nasdaq is by far the prevalent stock exchange in our database.

Regarding the event window, we initially employed a 15-second (10 intervals) window before and after a crash. However, we also conducted regressions on 12-second (8 intervals) and 18-second (12 intervals) windows. Regardless of the duration of the event window, our findings remained consistent.

Conclusion

Our study investigates the relationship between HFTs and ultra-fast flash events, challenging the methodologies employed in the academic and practitioner literature for identifying sudden liquidity black holes.

We question the criteria used for categorizing MFCs based on consecutive tick movements and percentage price changes. In particular, we question the requirements of 10 consecutive tick movements in the same direction of a crash and a price change of 0.8%. This practically means, for example, that a price change of 90% happening in less than 10 consecutive tick movements in the same direction would not qualify as an MFC, while a price change of 1% in 10 consecutive ticks would.

We also critique the accuracy of estimating EPMs by relying on predetermined fixed-time intervals, which may overlook important market dynamics. EPMs typically underestimate the magnitude of flash events because they rely on predetermined fixed-time intervals for identifying a shock (mechanically starting at the open of the market). This implies that price resiliency may already be occurring within the time interval of the EPM or that the shock may continue and impact the next time interval.

To refine the price-shock identification methodology, we introduce a new approach called SFEs, which focuses on sequences rather than fixed intervals. We pinpoint the exact time and duration of a crash and then build a window around it. As long as the tick curve does not reverse, we stay in the same sequence irrespective of the length of the time interval. These SFEs better capture the full extent of the shocks occurring in the market. Furthermore, we compare our sample of SFEs with those of MFCs and EPMs, employing close and open prices (OC), which replicates the original methodology. However, we also suggest using high and low (HL) prices, which better capture the extent of a shock within a predetermined time interval.

Our event studies around SFEs do not indicate that HFTs contribute to price instability. Since these event studies do not account for alternative trading and liquidity drivers such as relative spread, volatility, or volume, we conduct a multivariate regression analysis and reveal that HFTs increase their liquidity demand, on aggregate, during extreme price shocks and trade in the direction of a crash when it occurs. Otherwise, in the absence of shocks, they provide more liquidity than they absorb when spreads widen. Finally, our logit model indicates that SFEs are not related to HFTs consuming

more liquidity than they provide during the preceding time interval. Thus, we find no evidence that HFTs trigger extreme price shocks.

Our identification methodology based on SFEs is more accurate and leads to more robust empirical results overall. As they sometimes diverge from those obtained through MFCs or EPMs, we emphasize that conclusions regarding HFTs' contribution to price stability are contingent on the chosen methodology for identifying price shocks.

We, therefore, urge regulators to refrain from reaching premature conclusions about the role played by HFTs in stressful market conditions. Our findings show that the choice of the price-shock identification methodology is critical, and further research is necessary to test various identification methodologies on an expanded dataset. Stock exchanges should provide the academic community with such a database free of charge to promote greater transparency, while regulators should ensure that such a dataset is made available to the academic community.

This updated dataset should include at least one of the desirable features that we identify below. These suggestions for future research aim to address the gaps identified in the current research and to expand on the findings uncovered. The objective is not only to enhance the depth and breadth of understanding in this field but also to pave the way for future investigations.

First, the focus of this study is on HFTs and flash events in the U.S. market, which may present unique characteristics when compared to other markets. As of the time of this study, to the best of our knowledge, there appear to be no comparable datasets available that include a flag on HFT transactions and cover the same time period in Europe and Asia. When available, they could provide valuable insights and enhance the scope of our analysis.

Second, the Nasdaq HFT dataset we use in this study comprises aggregated data, so that we cannot have a granular view of the liquidity supply and demand at the HFT-firm level. Consequently, our findings are relevant at the HFT-group level only (which is still very important), but not at the HFT-firm level, which would provide further insight.

Third, large integrated firms (which may act as HFTs but not always) as well as HFTs routing their orders through large integrated firms have not been included by Nasdaq in the database due to the exchange's inability to precisely identify them. As such, the 26 HFT firms of the database can be considered as pure HFT firms or simply as "independent proprietary trading firms", as mentioned in Brogaard et al. (2017). The inclusion of data from larger integrated firms and HFTs using their services would undoubtedly enhance the breadth and accuracy of all previous studies, providing a fuller picture of the HFT landscape.

Table 11 Sample stocks summary statistics (Appendix)

Ticker	Company Name	Market Cap	Sector (GICS)
AA	Alcoa Corp	Large	Materials
AAPL	Apple Inc	Large	Information Technology
ABD	Acco Brands Corp	Small	Industrials
ADBE	Adobe Systems Inc	Large	Information Technology
AMED	Amedisys Inc	Medium	Health Care
AMGN	Amgen Inc	Large	Health Care
AMZN	Amazon.com Inc	Large	Consumer Discretionary
ARCC	Ares Capital Corp	Medium	Financials
AXP	American Express Co	Large	Financials
AYI	Acuity Brands Inc	Medium	Industrials
AZZ	AZZ Inc	Small	Industrials
BARE	Bare Escentuals Inc	Medium	Consumer Staples
BAS	Basic Energy Services Inc	Small	Energy
BHI	Baker Hughes a GE Co LLC	Large	Energy
BIIB	Biogen Inc	Large	Health Care
BRE	BEX Portfolio LLC	Medium	Real Estate
BXS	Bancorp South Inc	Medium	Financials
CBEY	Cbeyond Inc	Small	Telecommunications Services
CETV	Central European Media Enterprises Ltd	Medium	Consumer Discretionary
CNQR	Concur Technologies Inc	Medium	Information Technology
COO	Cooper Cos IncThe	Medium	Health Care
COST	Costco Wholesale Corp	Large	Consumer Staples
CR	Crane Co	Medium	Industrials
CSCO	Cisco Systems Inc	Large	Information Technology
CSE	Capital Source Inc	Medium	Financials
CSL	Carlisle Cos. Inc	Large	Industrials
CTSH	Cognizant Technology Solutions Corp	Large	Information Technology
DIS	Walt Disney CoThe	Large	Consumer Discretionary
DOW	Dow Chemical CoThe	Large	Materials
EBAY	eBay Inc	Large	Information Technology
ESRX	Express Scripts Holding Co	Large	Health Care
FCN	FTI Consulting Inc	Medium	Industrials
FL	Foot Locker Inc	Medium	Consumer Discretionary
FMER	FirstMerit Corp	Medium	Financials
FPO	First Potomac Realty Trust	Small	Real Estate
FULT	Fulton Financial Corp	Medium	Financials
GE	General Electric Co	Large	Industrials
GILD	Gilead Sciences Inc	Large	Health Care
GLW	Corning Inc	Large	Information Technology
GOOG	Alphabet Inc	Large	Information Technology
GPS	Gap IncThe	Large	Consumer Discretionary
HON	Honeywell International Inc	Large	Industrials
HPQ	HP Inc	Large	Information Technology
IMGN	ImmunoGen Inc	Small	Health Care
INTC	Intel Corp	Large	Information Technology
ISRG	Intuitive Surgical Inc	Large	Health Care
JKHY	Jack Henry & Associates Inc	Medium	Information Technology
KMB	Kimberly-Clark Corp	Large	Consumer Staples
KNOL	Knology Inc	Small	Telecommunications Services

Table 11 (continued)

Ticker	Company Name	Market Cap	Sector (GICS)
KR	Kroger CoThe	Large	Consumer Staples
LANC	Lancaster Colony Corp	Medium	Consumer Staples
LECO	Lincoln Electric Holdings Inc	Medium	Industrials
LPNT	LifePoint Health Inc	Medium	Health Care
LSTR	Landstar System Inc	Medium	Industrials
MANT	ManTech International CorpThe	Medium	Information Technology
MDCO	Medicines CoThe	Small	Health Care
MELI	Mercadorlibre Inc	Medium	Information Technology
MFB	Maidenform Brands LLC	Small	Consumer Discretionary
MMM	3M Co	Large	Industrials
MOD	Modine Manufacturing Co	Small	Consumer Discretionary
MOS	Mosaic CoThe	Large	Materials
NSR	NewStar Inc	Medium	Information Technology
NUS	Nu Skin Enterprises Inc	Medium	Consumer Staples
PFE	Pfizer Inc	Large	Health Care
PG	Procter & Gamble CoThe	Large	Consumer Staples
PNC	PNC Financial Services Group IncThe	Large	Financials
PNY	Piedmont National Gas Co Inc	Medium	Utilities
PTP	Platinum Underwriters Holdings Ltd	Medium	Financials
RIGL	Rigel Pharmaceuticals Inc	Small	Health Care
ROC	Rockwood Holdings Inc	Medium	Materials
ROCK	Gibraltar Industries Inc	Medium	Industrials
SF	Stifel Financial Corp	Medium	Financials
SFG	StanCorp Financial Group Inc	Medium	Financials
SWN	Southwestern Energy Co	Large	Energy

The table reports the ticker, company name, market cap, sector (GICS classification) regarding the 74 stocks of our sample. Market capitalization data are provided by Bloomberg. The group classification is the same whether we use Bloomberg's historical market capitalization as of February 26, 2010, Bloomberg's 2008, 2009, 2010 historical market capitalizations or Bloomberg's 2008, 2009, 2010 average historical market capitalization

Table 12 Comparison of the different methodologies based on empirical results

Panel A : HFT liquidity supply and demand around ultra-fast flash events						
Methodology	Variable	$t - 2$	$t - 1$	t	$t + 1$	$t + 2$
MFCs	HFT^{NET}	NS	NS	NS	NS	NS
SFEs	HFT^{NET}	—	—	—	—	—
OC-EPMs	HFT^{NET}	++	NS	+++	+++	++
HL-EPMs	HFT^{NET}	NS	NS	NS	+++	NS
Panel B : Multivariate regressions - Net HFT imbalance during ultra-fast flash events						
Methodology	Equations	All	Standalone	Simultaneous	Extreme-Hours	
MFCs	5.1, 5.2, 5.3	—	—	NS	NS	
SFEs	5.1, 5.2, 5.3	+++	+++	+++	NS	
OC-EPMs	5.1, 5.2, 5.3	NS	NS	NS	—	
HL-EPMs	5.1, 5.2, 5.3	NS	NS	NS	NS	
Panel C : Logit - Ultra-fast flash events						
Methodology	Variable	All	Standalone	Simultaneous	Extreme-Hours	
MFCs	HFT_{it-1}^{NET}	NS	NS	NS	NS	
SFEs	HFT_{it-1}^{NET}	—	—	NS	NS	
OC-EPMs	HFT_{it-1}^{NET}	—	—	—	—	
HL-EPMs	HFT_{it-1}^{NET}	NS	—	—	—	

The table highlights the differences between the new methodology (SFEs) and the traditional methodologies (MFCs, EPMs). NS stands for not significant, — stands for negative at the 5% level, — stands for negative at the 1% level, ++ stands for positive at the 5% level, and +++ stands for positive at the 1% level

Fourth, the dataset is limited to transactions executed on Nasdaq, which means that HFT transactions executed on other U.S. stock exchanges are not considered. Although we account for this by filtering events based on the proportion of trades executed on Nasdaq, this approach inevitably results in a partial view. The potential impact of HFTs (or NHFTs) operating outside Nasdaq, particularly in instances where they may trigger market fluctuations or crashes, remains unobserved. Therefore, integrating data from these other exchanges would enable a more holistic analysis, offering deeper insights into the dynamics of HFT activities across the broader U.S. stock-market landscape.

Finally, we cannot control for market manipulation or the potential imbalance generated by NHFT algorithmic traders. To do so, we would need more granular data with an extra flag on NHFT algorithmic trades, which are not provided by Nasdaq.

Appendix

See appendix Tables 11 and 12.

Abbreviations

AA	Alcoa
AAPL	Apple
DMM	Designated market maker
EPM	Extreme price movement
HFT	High-frequency trader
HL-EPM	Extreme price movement based on high and low prices
MFC	Mini flash crash
NBBO	National best bid and offer
NHFT	Non-high frequency trader
OC-EPM	Extreme price movement based on open and close prices

SFE Sequence-based flash event

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Author contributions

Christophe DESAGRE prepared the dataset, wrote the algos, conducted statistical and econometric tests, and reviewed the paper. Floris LALY extracted data, wrote the draft of the paper, conducted robustness checks, and re-viewed the paper. Mikael PETITJEAN defined the methodology, interpreted the results, and finalized the paper.

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Data availability

The dataset that supports the findings of this study is the so called Nasdaq HFT dataset, which are available from NASDAQ OMX but restrictions apply to the availability of these data, which were used under a non-disclosure agreement signed by the authors and Frank Hatheway from NASDAQ OMX. Data may be available from the authors upon reasonable request and with permission of NASDAQ OMX.

Declarations

Competing interests

In the spirit of transparency and to maintain the integrity of our research, we declare that we have no competing interests that could potentially influence the outcomes or interpretations pre-sented in this research paper. We confirm that there are no financial, personal, or professional relationships that might be perceived as creating conflicts of interests with regard to the research conducted. This includes, but is not limited to, financial support, grants, employment, consultancies, honoraria, patents, stock ownership, or any other relationships that could bias the findings or conclusions of this study. We have conducted this research with the utmost objectivity and have strived to ensure that the information provided is accurate, unbiased, and in the best interest of advancing scientific knowledge.

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References

- Abreu D (2022) 'High frequency traders and liquidity', Working paper
- Aldrich E, Grundfest J, Laughlin G (2017) 'The flash crash: a new deconstruction', *Working paper*
- Anand A, Venkataraman K (2016) Market conditions, fragility, and the economics of market making. *J Financ Econ* 121(2):327–349
- Aquilina M, Eyles B, Shao J, Ysusi C (2018) How do participants behave during flash events? evidence from the UK equity market. Financial Conduct Authority Research Note, London
- Bellia M, Christensen K, Kolokolov A, Pelizzon L, Renò R (2022) 'High-frequency trading during flash crashes: walk of fame or hall of shame?', *Working paper*
- Bessembinder H, Hao J, Zheng K (2020) Liquidity provision contracts and market quality: Evidence from the new york stock exchange. *Rev Financ Stud* 33(1):44–74
- Boehmer E, Fong K, Wu JJ (2021) Algorithmic trading and market quality: international evidence. *J Financ and Quant Anal* 56(8):2659–2688
- Boehmer E, Li D, Saar G (2018) The competitive landscape of high-frequency trading firms. *Rev Financ Stud* 31(6):2227–2276
- Braun T, Fiegen JA, Wagner DC, Krause SM, Guhr T (2018) Impact and recovery process of mini flash crashes: an empirical study. *PLoS ONE* 13(5):e0196920
- Breckenfelder J (2019) 'Competition among high-frequency traders, and market quality', ECB Working Paper No. 2290.
- Brogaard J, Carrion A, Moyaert T, Riordan R, Shkillo A, Sokolov K (2018) High frequency trading and extreme price movements. *J Financ Econ* 128(2):253–265
- Brogaard J, Hendershott T, Riordan R (2014) High-frequency trading and price discovery. *Rev Financ Stud* 27(8):2267–2306
- Brogaard J, Hendershott T, Riordan R (2017) High frequency trading and the 2008 short-sale ban. *J Financ Econ* 124(1):22–42
- Carrion A (2013) Very fast money: high-frequency trading on the nasdaq. *J Financ Mark* 16(4):680–711
- Christensen K, Oomen R, Renò R (2022) The drift burst hypothesis. *J Econ* 227(2):461–497

- Clark-Joseph AD, Ye M, Zi C (2017) Designated market makers still matter: Evidence from two natural experiments. *J Financ Econ* 126(3):652–667
- Demirer R, Leggio KB, Lien D (2019) 'Herdin and flash events: evidence from the 2010 flash crash', *Finance Research Letters* 31
- Félez-Vinas E (2018) Market fragmentation, mini flash crashes and liquidity, Working paper presented at the FMA European Conference, Kristiansand, Norway
- Firth D (1993) Bias reduction of maximum likelihood estimates. *Biometrika* 80(1):27–38
- Gao C, Mizrach B (2016) Market quality breakdowns in equities. *J Financ Mark* 28:1–23
- Gerig A (2015) 'High-frequency trading synchronizes prices in financial markets', *Working paper*
- Goldstein MA, Kumar P, Graves FC (2014) Computerized and high-frequency trading. *Financ Rev* 49(2):177–202
- Goldstein M, Kwan A, Philip R (2023) High-frequency trading strategies. *Manag Sci* 69(8):4413–4434
- Golub A, Keane J, Poon S-H (2012) 'High frequency trading and mini flash crashes', *Working paper*
- Hagströmer B, Nordén L (2013) The diversity of high-frequency traders. *J Financ Mark* 16(4):741–770
- Hasbrouck J, Saar G (2013) Low-latency trading. *J Financ Mark* 16(4):646–679
- Hirschey N (2021) Do high-frequency traders anticipate buying and selling pressure? *Manag Sci* 67(6):3321–3345
- Johnson N, Zhao G, Hunsader E, Qi H, Johnson N, Meng J, Tivnan B (2013) Abrupt rise of new machine ecology beyond human response time. *Sci Rep* 3:2627
- Jones CM (2013) 'What do we know about high-frequency trading?', *Columbia Business School Research Paper* (13-11)
- Kirilenko A, Kyle AS, Samadi M, Tuzun T (2017) The flash crash: high-frequency trading in an electronic market. *J Financ* 72(3):967–998
- Laly F, Petitjean M (2020) Mini flash crashes: review, taxonomy and policy responses. *Bull Econ Res* 72(3):251–271
- Leal SJ, Napoletano M, Roventini A, Fagiolo G (2016) Rock around the clock: an agent-based model of low-and high-frequency trading. *J Evol Econ* 26(1):49–76
- Longin FM (2000) From value at risk to stress testing: the extreme value approach. *J Bank & Financ* 24(7):1097–1130
- Massad T (2015) 'Remarks of chairman timothy massad before the conference on the evolving structure of the U.S. treasury market'
- Menkveld AJ (2014) High-frequency traders and market structure. *Financ Rev* 49(2):333–344
- Menkveld AJ, Yueshen BZ (2019) The flash crash: a cautionary tale about highly fragmented markets. *Manag Sci* 65(10):4470–4488
- Menkveld AJ, Zoican MA (2017) Need for speed? exchange latency and liquidity. *Rev Financ Stud* 30(4):1188–1228
- Nanex (2010) 'Flash equity failures in 2006, 2007, 2008, 2009, 2010, and 2011, technical report'
- O'Hara M, Yao C, Ye M (2014) What's not there: odd lots and market data. *J Financ* 69:2199–2236
- Ozenbas D, Schwartz RA (2018) Do high frequency trading firms provide two-sided liquidity? *J Portf Manag* 44(7):63–74
- Scholtus M, van Dijk D, Frijns B (2014) Speed, algorithmic trading, and market quality around macroeconomic news announcements. *J Bank & Financ* 38:89–105
- Shkilkov A, Sokolov K (2020) Every cloud has a silver lining: fast trading, microwave connectivity, and trading costs. *J Financ* 75(6):2899–2927
- Steffen V (2023) 'A literature review on extreme price movements with reversal', *Journal of Behavioral and Experimental Finance* p. 100806
- Stoll HR (2014) High speed equities trading: 1993–2012. *Asia Pac J Financ Stud* 43(6):767–797
- Zaharudin KZ, Young MR, Hsu W-H (2022) High-frequency trading: Definition, implications, and controversies. *J Econ Surv* 36(1):75–107

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